Syracuse University

SAL 603: Final Project

A cross-league analysis of offensive and defensive metrics and win percentage

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Research Questions

- ✓ Which offensive and defensive performance metrics are most strongly correlated with team win percentage across professional sports leagues (NFL, NBA, MLB, NHL, EPL)?
- ✓ Is team success more strongly associated with peak single-season statistical output or with consistent performance sustained across multiple seasons?

NFL OVERVIEW

All Data Gathered from [Pro-Football Reference]

NFL Data Source

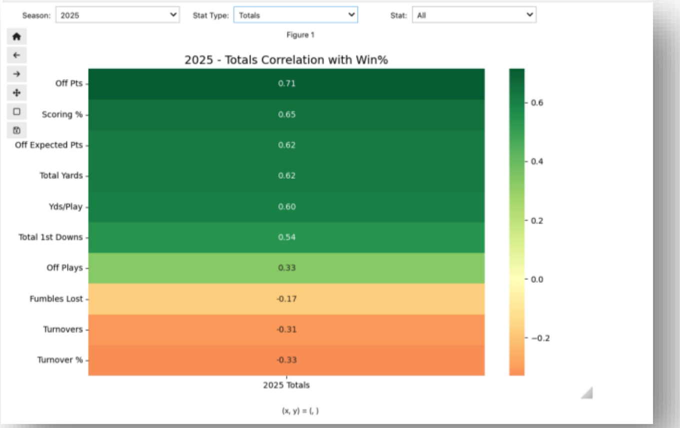
```
[4]: #Started with NFL offensive stats, renamed a column 'PF': 'Pts', added two columns, stripped white space, and eliminated an unnamed column
     #that was included from the CSV. Added in Off_ at the beginning of each column name, so when the data was merged with defensive stats
     #Offensive and defensive columns would be easier to identify
     #Also dealt with the double row header using (header=[0,1])

     nfl_2025_team_offensive_stats = pd.read_csv("Team_Offensive_Stats_2025.csv", header=[0,1])
     nfl_2025_team_offensive_stats.rename(columns={'PF': 'Pts'}, inplace=True)
     nfl_2025_team_offensive_stats ["Stat Type"] = "Offensive"
     nfl_2025_team_offensive_stats ["Season"] = "2025"
     nfl_2025_team_offensive_stats.columns = [
         f"{top}_{bottom}".strip('_') if top != bottom and 'Unnamed' not in str(top)
         else bottom
         for top, bottom in nfl_2025_team_offensive_stats.columns
     ]
     nfl_2025_team_offensive_stats.columns = [f"off_{col}" if col not in ['Rk', 'Tm', 'G'] else col for col in nfl_2025_team_offensive_stats.colum

[5]: #Imported team records for the 2025 season, stripped out white space and removed footnotes from team names

     nfl_2025_team_records = pd.read_csv("NFL_Team_Records_2025.csv")
     nfl_2025_team_records.columns = nfl_2025_team_records.columns.str.strip()
     nfl_2025_team_records['Tm'] = nfl_2025_team_records['Tm'].str.replace(r'[*#+]', '', regex=True).str.strip()

[6]: #Combined the 2025 offensive states with the team records from 2025 using a left-hand merge on Tm

     nfl_2025_team_stats_records_combined = nfl_2025_team_records.merge(nfl_2025_team_offensive_stats, on="Tm", how="left")

[7]: #Follow the same steps as with the 2025 season data above.

     nfl_2024_team_offensive_stats = pd.read_csv("Team_Offensive_Stats_2024.csv", header=[0,1])
     nfl_2024_team_offensive_stats ["Stat Type"] = "Offensive"
     nfl_2024_team_offensive_stats ["Season"] = "2024"
     nfl_2024_team_offensive_stats.columns = [
         f"{top}_{bottom}".strip('_') if top != bottom and 'Unnamed' not in str(top)
         else bottom
         for top, bottom in nfl_2024_team_offensive_stats.columns
     ]
     nfl_2024_team_offensive_stats.columns = [f"off_{col}" if col not in ['Rk', 'Tm', 'G'] else col for col in nfl_2024_team_offensive_stats.colum
     nfl_2024_team_offensive_stats = nfl_2024_team_offensive_stats.set_index("Tm")

[8]: #Follow the same steps as with the 2025 season data above.

     nfl_2024_team_records = pd.read_csv("NFL_Team_Records_2024.csv")
     nfl_2024_team_records.columns = nfl_2024_team_records.columns.str.strip()
     nfl_2024_team_records['Tm'] = nfl_2024_team_records['Tm'].str.replace(r'[*#+]', '', regex=True).str.strip()
```

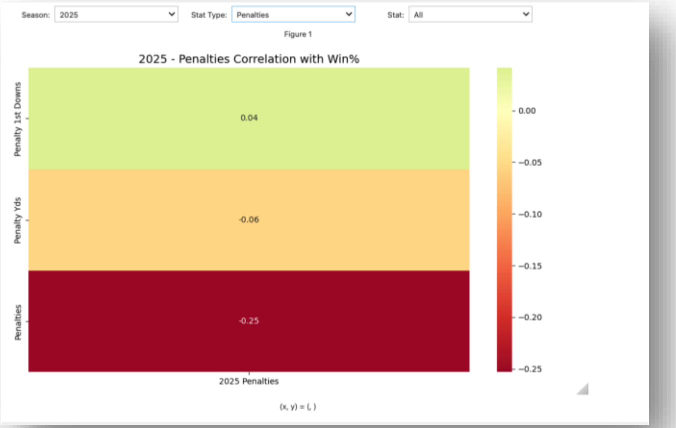
Data Source

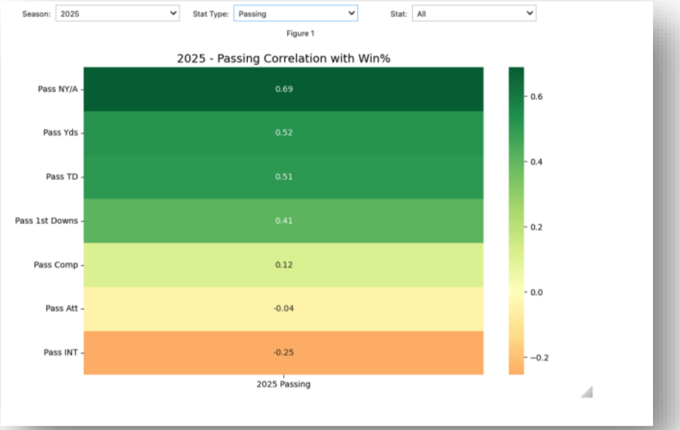
Overview:

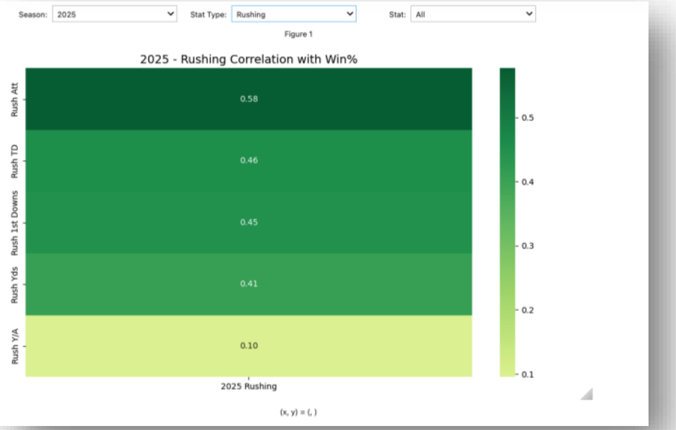
- All raw data for the NFL section was sourced from Pro Football Reference.
- Information was exported into an Excel file and then converted into a CSV file that could be passed into a Jupyter Notebook in the Anaconda Navigator Environment using Python.
- Team Records, Offensive Statistics and Defensive statistics were collected and utilized for this assignment.
- Claude was used to help implement dashboards.

NFL Offensive Correlations: 2025 Season









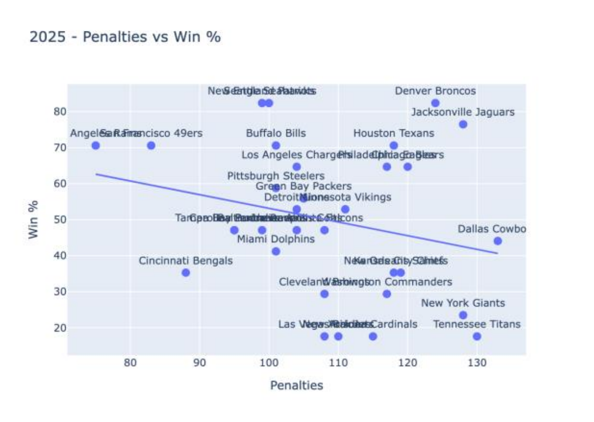
Seasons Reviewed: 2021—2025

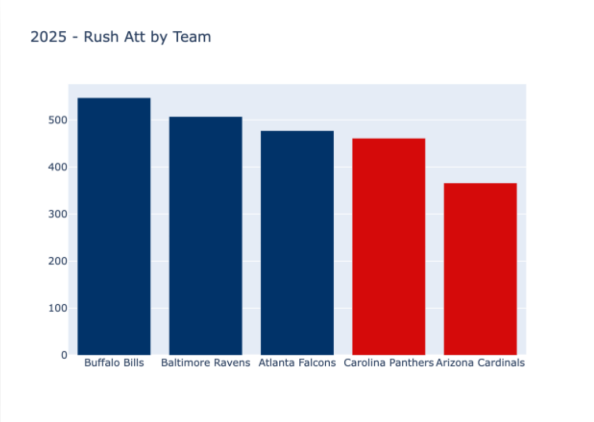
Findings:

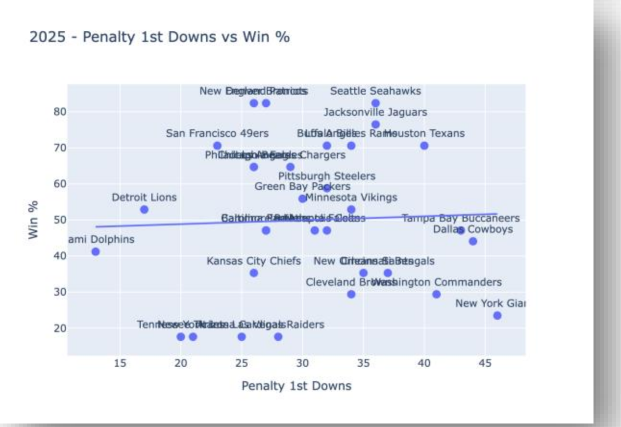
- Offensive totals (top left) had the strongest correlation with win percentage
- Rushing statistics didn't possess the high-end upside of passing statistics, but were much more consistent
- As expected, ***passing interceptions*** have a negative correlation with win percentage and expose the risk of high-volume passing offenses
- The ***number of penalties*** had a higher negative correlation percentage than ***total penalty yards***
- ***Interceptions*** had a larger negative correlation than fumbles lost

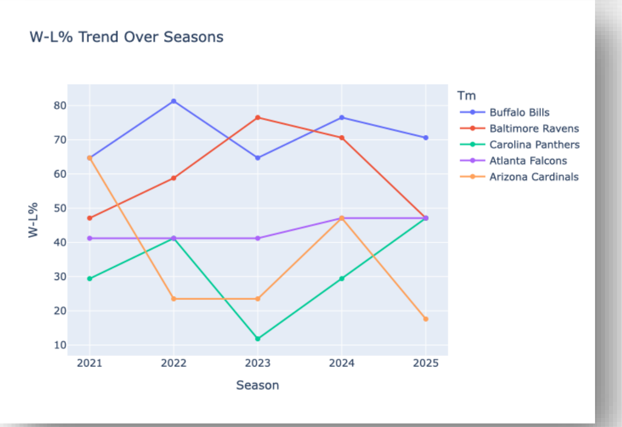
Based on 2025 data shown in graphics, other seasons may have some variance

NFL Offensive Correlations: 2025 Season









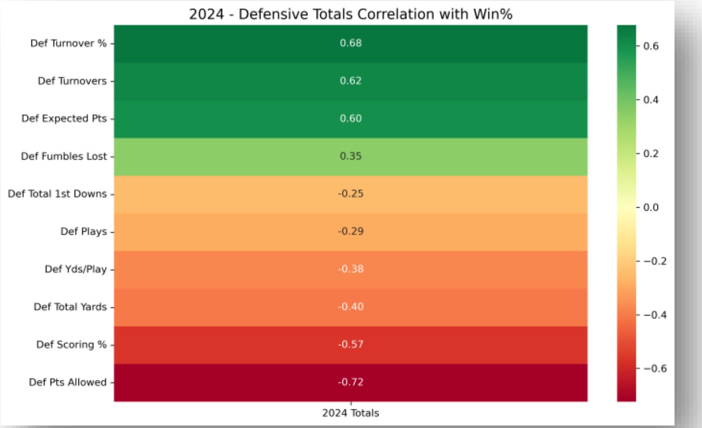
Seasons Reviewed: 2021—2025

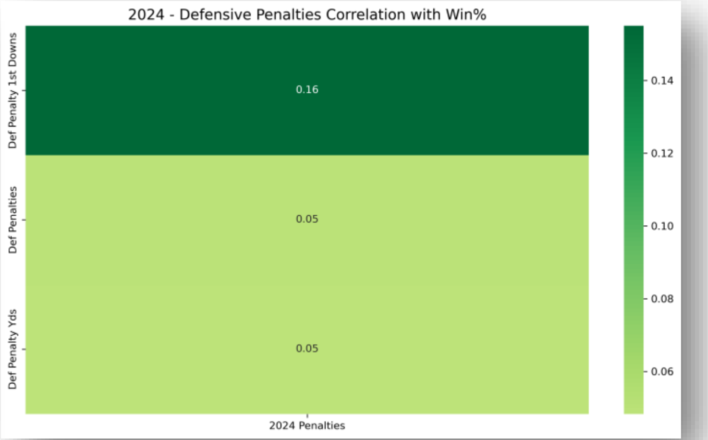
Overview:

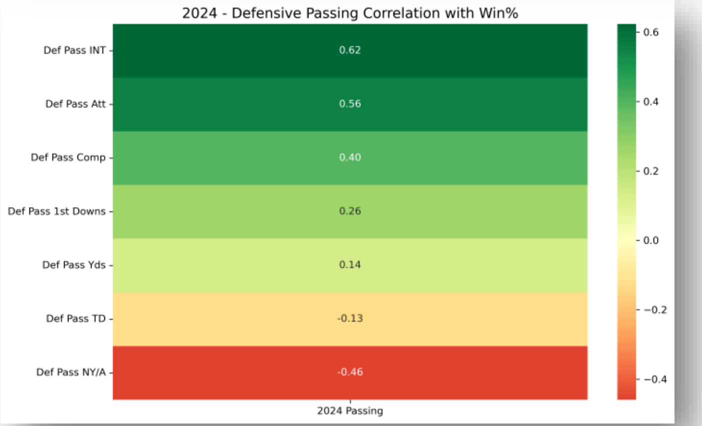
- Our team develops several dashboards within our Python environment that allow a user to toggle between statistics to enhance the experience of interacting with the dataset.
- As seen in the pictures, the user can see how different offensive statistics correlate with win percentage and how each team performed.
- Additionally, we developed a team compare tab to allow users to view statistical output side by side by team.
- Finally, there is a trend analysis tab for the user to see how a team has performed over the seasons contained in the dataset.

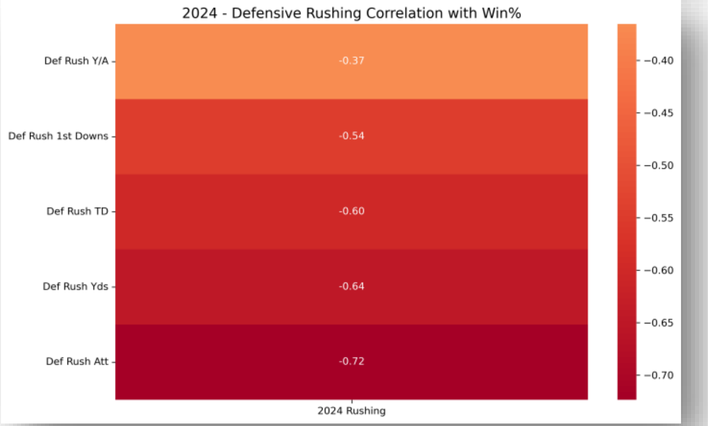
Based on 2025 data shown in graphics, other seasons may have some variance

NFL Defensive Correlations: 2024









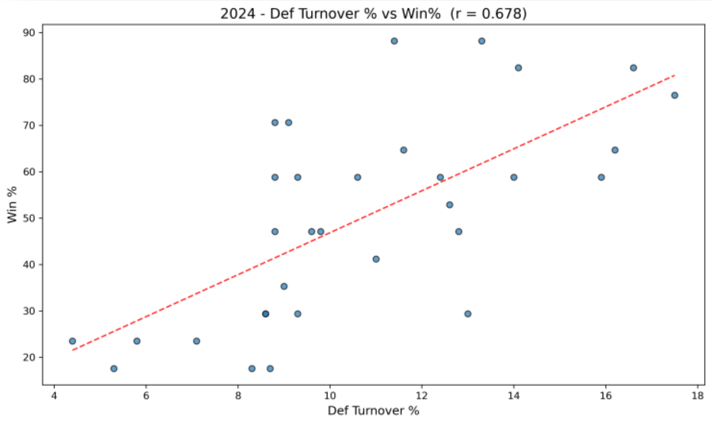
Seasons Reviewed: 2021—2025

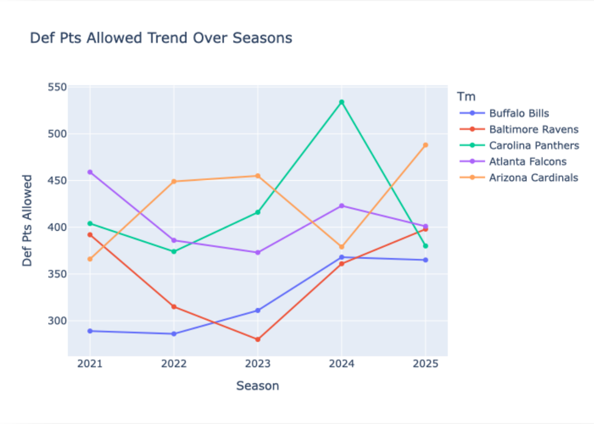
Findings:

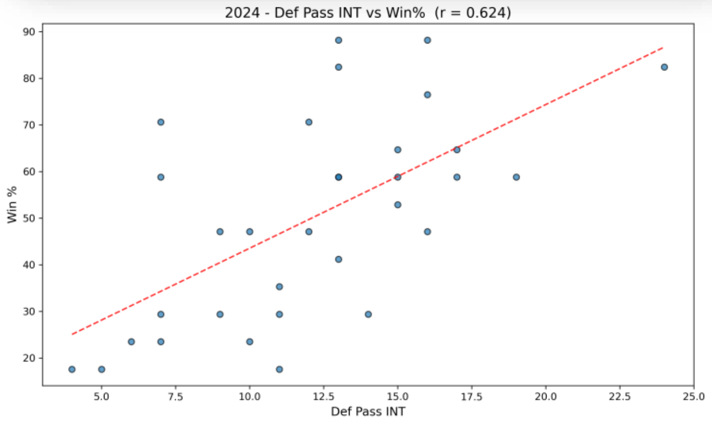
- Defenses that cause a high ***turnover rate*** positively correlate with win percentage.
- ***Rushing stats*** given up by a defense statistical category have a high negative correlation with win percentage. This likely supports the notion that controlling the line of scrimmage is essential to consistently winning football games.
- Coaches and front offices may want to increase focus on ***defensive net yards per attempt***, as it has the highest negative correlation with win percentage (based on 2024 data shown in the image).
- ***Defensive penalties*** have minimal effect on win percentage compared to the other statistics reviewed.

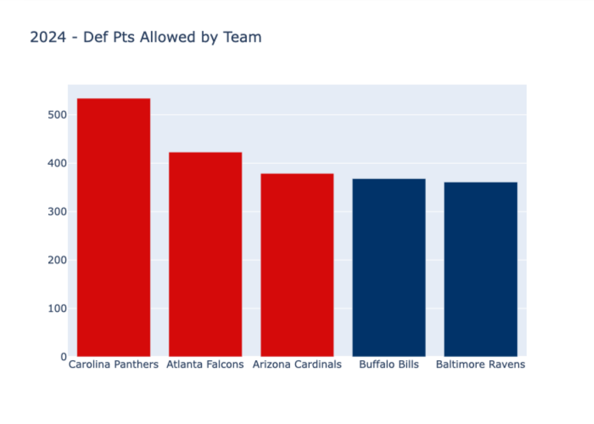
Based on 2024 data shown in graphics, other seasons may have some variance

NFL Defensive Correlations: 2024









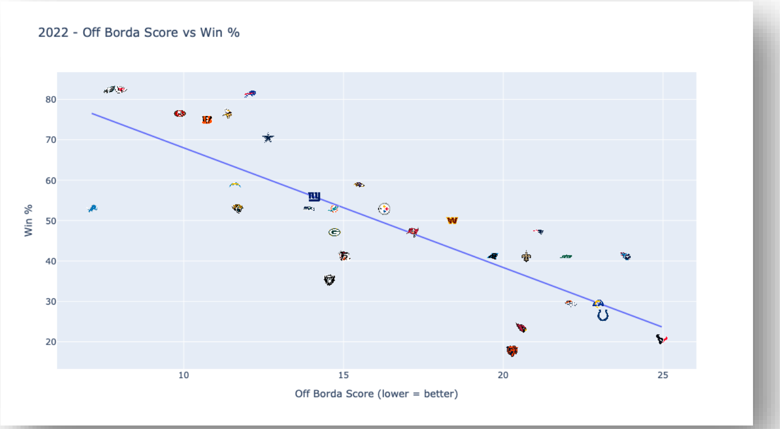
Seasons Reviewed: 2021—2025

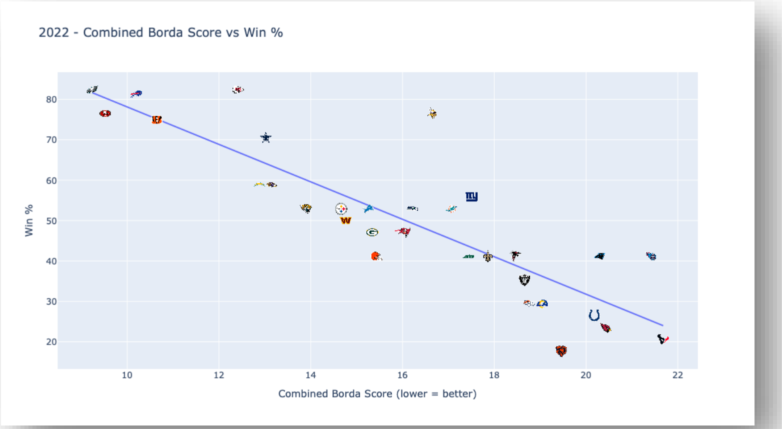
Overview: Our team developed defensive metrics dashboards in our Python environment that allow users to toggle between defensive statistics to enhance their experience interacting with the dataset.

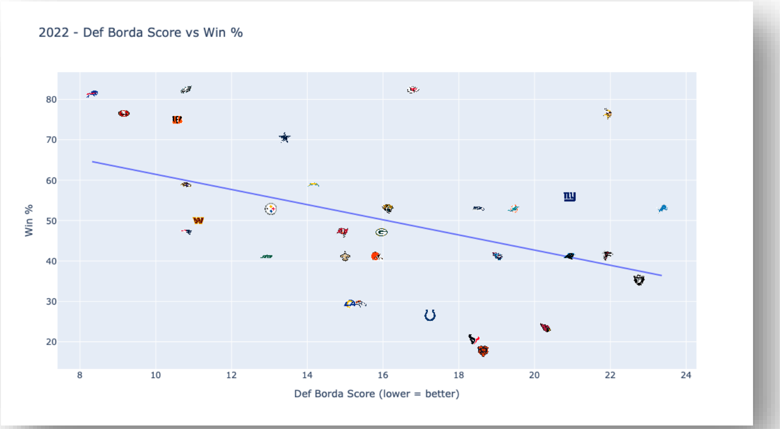
- As seen in the pictures, a user can observe how different defensive statistics correlated with win percentage and how each team performed.
- Again, we developed a team compare tab to let users have a side-by-side view of statistical output by team.
- Finally, there is a trend analysis tab for the user to see how a team has performed over the seasons contained in the dataset.

Based on 2025 data shown in picture, other seasons may have some variance

NFL Borda Correlations: 2022





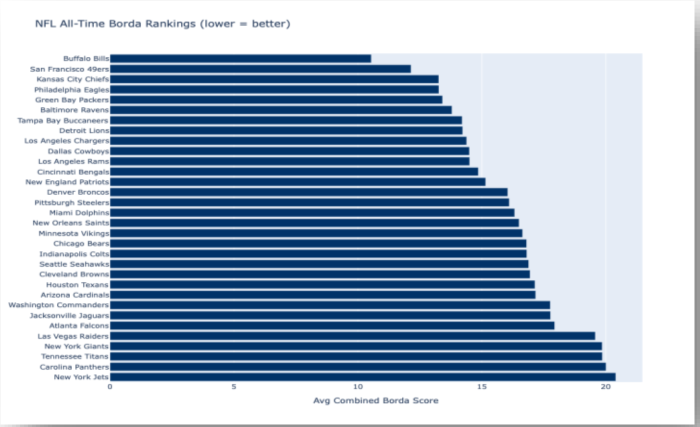


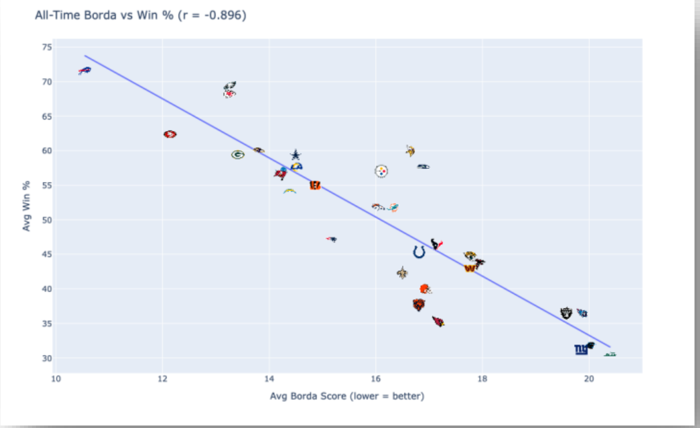
As our team progressed through the project, we became increasingly interested in applying Borda scores to our data sets across all five leagues being reviewed to eventually develop a power ranking based on both an aggregate composite statistic for the five seasons in the dataset and a consistency metric to see how consistent teams have been with their statistical performance over the past five seasons.

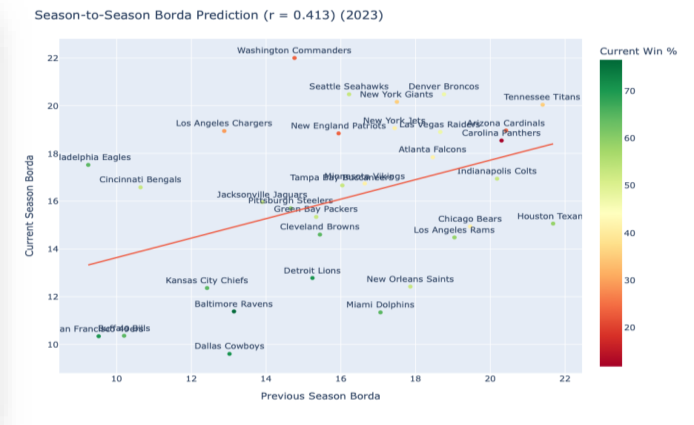
These are all contained within dashboard environments created in Python

One important point to note, which we discuss during our video presentation, is that a negative correlation is the expected result because a lower Borda score indicates better statistical performance. This goes against what is usually seen with sports data, so we felt it was necessary to emphasize that negative correlations are intentional.

NFL Borda Consistency Rankings





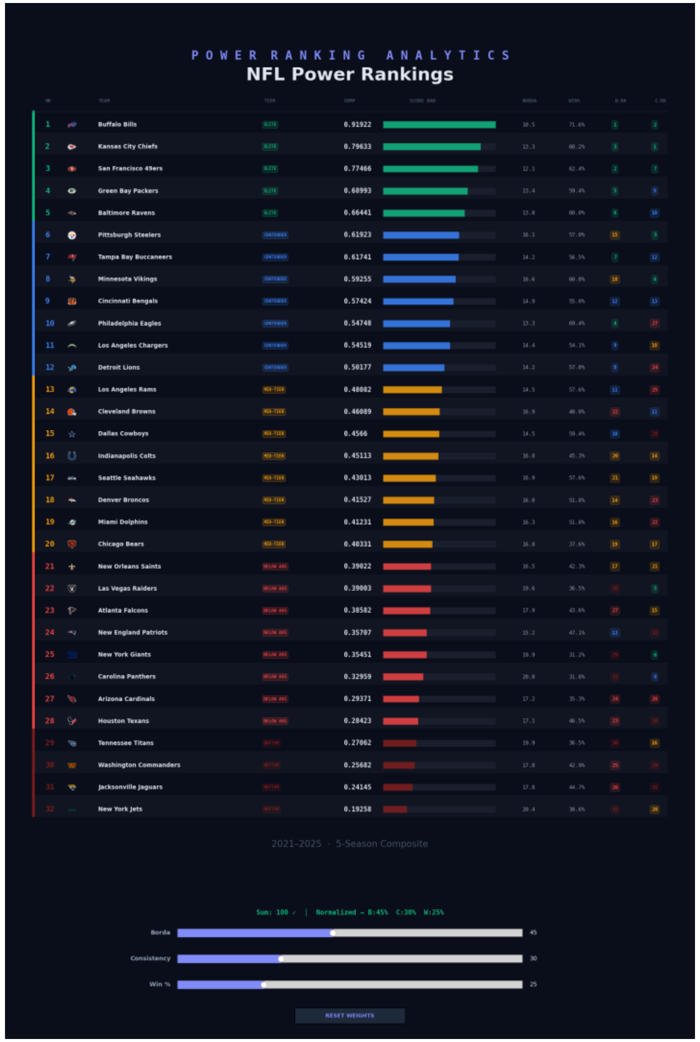


Seasons Reviewed: 2021—2025

Overview: Based on the Borda scores, we set out to determine inter-season consistency to see how teams performed over the period.

- The **Bills**, as seen in the graphic on the top left, have been the most consistent teams in terms of statistical performance (offense +defense) over the past 5 seasons.
- The correlation of the Borda scores, season-to-season, showed wide variability, but the 2023 season had the highest r at 0.413 (a moderate but notable finding given the noise inherent in NFL season-to-season performance).
- Finally, the quality vs consistently graphic aims to put teams into quadrants to help the end user visualize how teams have performed relative to their win percentage (size of the circle).
- Two interesting teams to note are the Philadelphia Eagles and the PittsburghSteelers (see next slide)

NFL Borda: Power Rankings



Seasons Reviewed: 2021—2025

Talking Points:

- **Pittsburgh Steelers:** Steelers fans have had significant frustration over the past several seasons because they feel the team is consistently average. The data set confirms these frustrations: the Steelers recorded a combined Borda of 15th but ranked 3rd in terms of inconsistency over the past 5 seasons.
- **Philadelphia Eagles:** During the seasons contained in the dataset, the Eagles have been one of the few teams to win a Super Bowl, and their combined Borda ranking confirms they have been elite in statistical output, ranking 4th; however, the team has also been wildly inconsistent.
- **Consistency doesn't equal success**: An interesting finding was that several of the lowest-rated Borda teams scored highly on season-to-season consistency rankings. We felt this was a very important finding. While correlations between win percentage and statistical output are vital, understanding when a franchise has plateaued, we felt, was equally important.
- **High Borda doesn't equal championship**: Finally, the Buffalo Bills had the highest-ranked combined Borda and ranked 2nd in consistency; however, during the past 5 seasons, the Bills have never made it to a Super Bowl. This led us to consider the Moneyball aspects of building a team that could consistently win in the regular season but lacked certain attributes when it mattered most.
- **Future Direction:** Include playoff data and determine the importance of defensive metrics as the talent gap narrows in the playoffs.

EPL OVERVIEW

All Data Gathered from FB-Reference (Soccer)

EPL Data Source

```
[49]: #Reset the column names to match the dictionaries above

      epl_own_team_stats_2024 = epl_own_team_stats_2024.rename(columns=epl_off_rename)
      epl_opp_team_stats_2024 = epl_opp_team_stats_2024.rename(columns=epl_def_rename)

[50]: #Read in the team records from the EPL season

      epl_team_records_2024 = pd.read_csv("EPL_Data/epl_team_records_2024.csv")
      epl_team_records_2024['Squad'] = epl_team_records_2024['Squad'].str.strip()

[51]: #Merged the offensive stats with the team records for the season, created a win % column and calculation, created a new season column, and di
      #Repeated these steps for the defensive statistics

      epl_2024_offensive = epl_team_records_2024.merge(epl_own_team_stats_2024, on='Squad', how='left')
      epl_2024_offensive['W_pct'] = (epl_2024_offensive.W + 0.5 * epl_2024_offensive.D) / (
          epl_2024_offensive.W + epl_2024_offensive.D + epl_2024_offensive.L)
      epl_2024_offensive ['Season'] = '2024'
      epl_2024_offensive = epl_2024_offensive.drop(columns=['Off Players Used'])

      epl_2024_defensive = epl_team_records_2024.copy().merge(epl_opp_team_stats_2024, on='Squad', how='left')
      epl_2024_defensive['W_pct'] = (epl_2024_defensive.W + 0.5 * epl_2024_defensive.D) / (
          epl_2024_defensive.W + epl_2024_defensive.D + epl_2024_defensive.L)
      epl_2024_defensive ['Season'] = '2024'
      epl_2024_defensive = epl_2024_defensive.drop(columns=['Opp Players Used'])

[52]: #Repeated the same steps from above for the EPL 2024 data for the EPL 2023 data
      #Read in each season of data individually to ensure data integrity

      epl_own_team_stats_2023 = pd.read_csv("EPL_Data/epl_own_team_stats_2023.csv", header=1)
      epl_own_team_stats_2023.columns = col_names
      epl_own_team_stats_2023.columns = [f"own_{c}" if c != 'Squad' else c for c in epl_own_team_stats_2023.columns]
      epl_own_team_stats_2023['Squad'] = epl_own_team_stats_2023['Squad'].str.strip()

[53]: #Repeated the same steps from above for the EPL 2024 data for the EPL 2023 data
      #Read in each season of data individually to ensure data integrity

      epl_team_records_2023 = pd.read_csv("EPL_Data/epl_team_records_2023.csv")
      epl_team_records_2023 = epl_team_records_2023.dropna(how='all')
      epl_team_records_2023['Squad'] = epl_team_records_2023['Squad'].str.strip()

[54]: #Repeated the same steps from above for the EPL 2024 data for the EPL 2023 data
      #Read in each season of data individually to ensure data integrity

      epl_opp_team_stats_2023 = pd.read_csv("EPL_Data/epl_opp_team_stats_2023.csv", header=1)
      epl_opp_team_stats_2023.columns = col_names
```

Data Source

Overview:

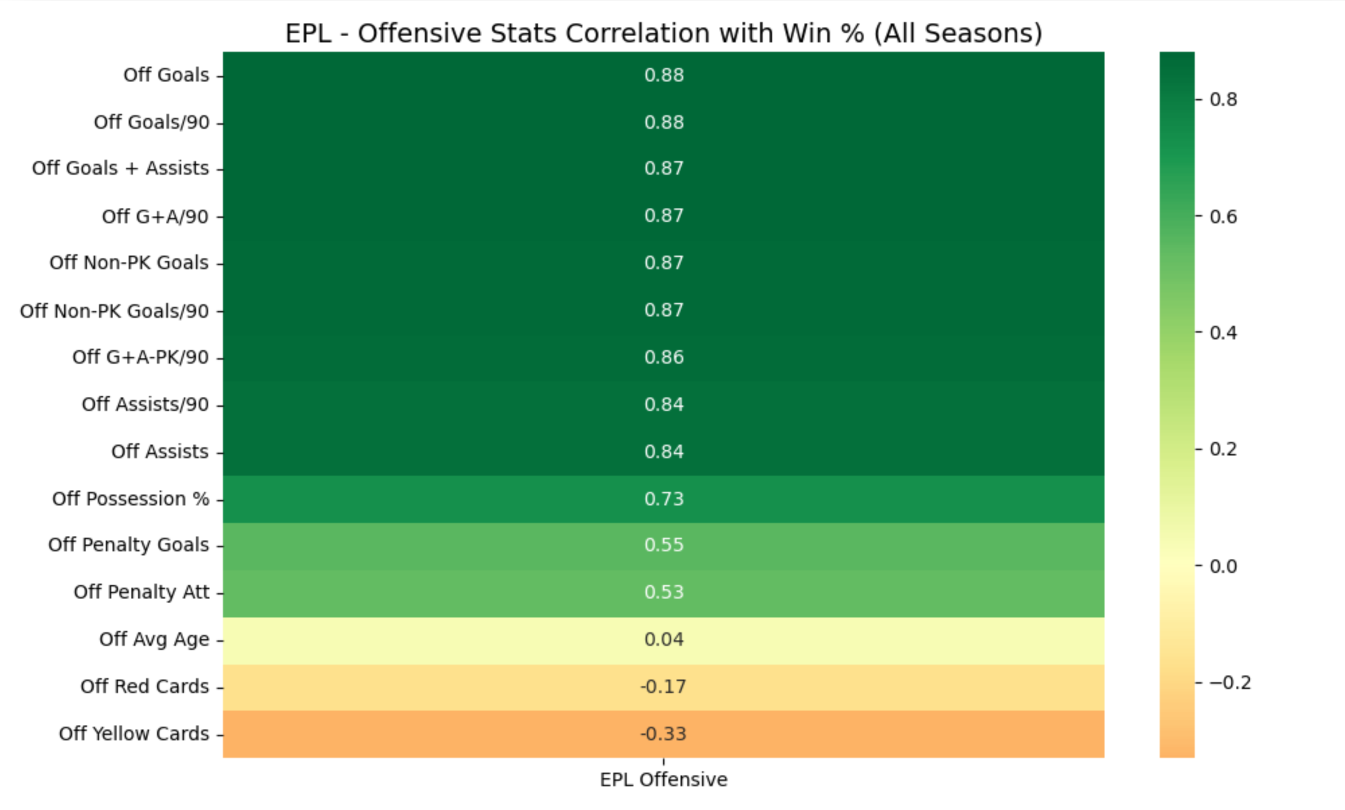
- All raw data for the EPL section was sourced from FB (Soccer) Reference.
- Information was exported into an Excel file and then converted into a CSV file that could be passed into a Jupyter Notebook in the Anaconda Navigator Environment using Python.
- Team Records, Offensive Statistics and Defensive statistics were collected and utilized for this assignment.
- Claude was used to help implement dashboards.

EPL vs American Sports League: Challenges

Challenges:

- Major American professional sports leagues do not have a promotion/relegation system, making direct cross-league comparisons with the EPL difficult
- American leagues feature 30+ teams with consistent season-to-season membership, whereas the EPL has only 20 teams with roster turnover due to relegation
- The structural differences affect how consistency metrics can be applied. EPL teams may not appear in the dataset across all 5 seasons.
- While not a focus of this review, the financial impact of relegation on lower-tier EPL teams introduces a competitive disadvantage that has no equivalent in American sports leagues.

EPL Offensive Correlations: All Seasons

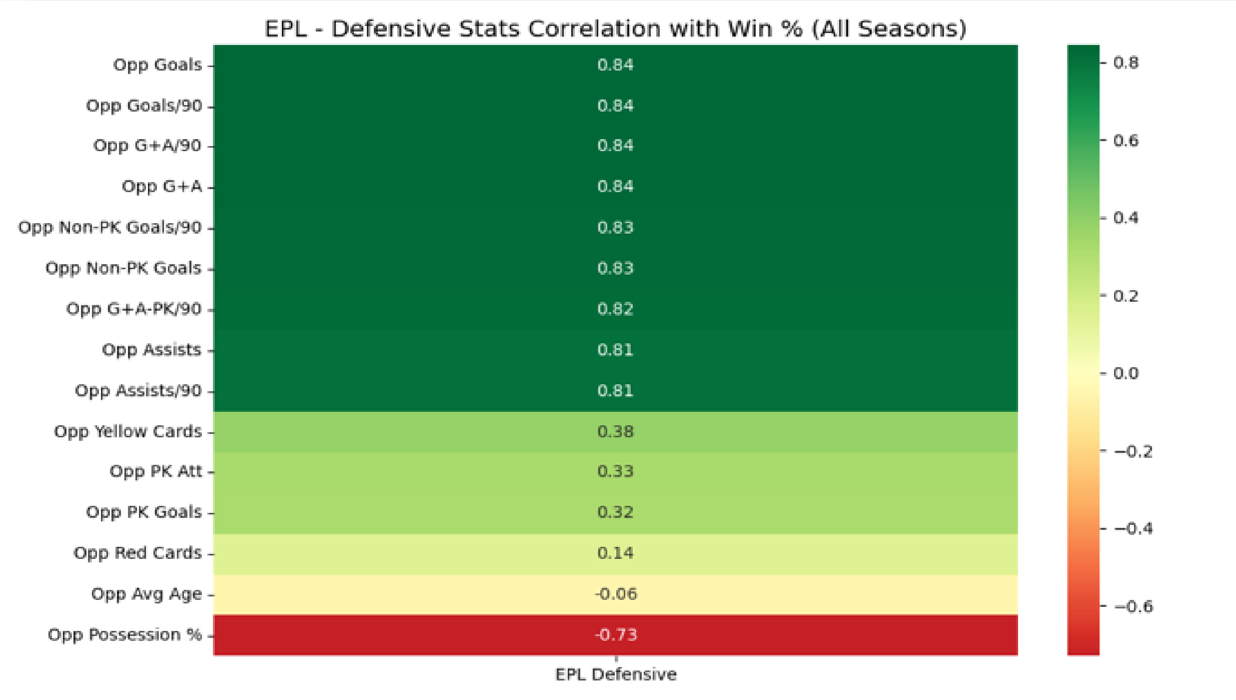


Seasons Reviews: 2020—2024

Findings:

- Based on running correlations between offensive EPL metric data against win percentage, the results returned what was largerly expect. Goals, Assists and Goals+Assists per 90 minutes were all highly correlated with win percentage.
- This allowed our team to have a good foundational point of reference as we built towards our Borda Score Power Ranking for the EPL.
- Additionally, we would it interesting that age was a neutral factor due to the desire for many EPL teams to always be looking to aquire young talent and sell on experience players

EPL Defensive Correlations: All Seasons

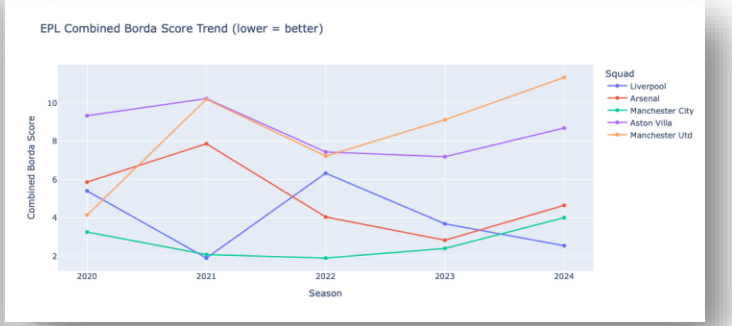


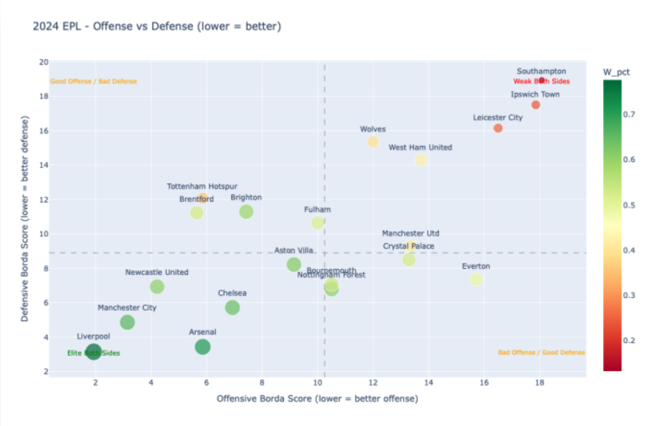
Seasons Reviewed: 2020—2024

Findings:
Defensive metrics also yielded some very intriguing findings, with opponent possession percentage showing a strong negative correlation with win percentage.

- This reinforces the possession-based strategy that many of the larger, more financially capable clubs have adopted over the past decade. The logic behind those tactics makes a lot of sense.
- However, having the players with the skills and ability to execute a strong possession-based footballing strategy is incredibly difficult.
- Although beyond the scope of this research, future studies could further explore financial disparities, shedding light on the gap between elite clubs and mid- to lower-tier teams. Such analyses might also clarify how possession-based football, winning percentage, and financial strength are strongly correlated.

EPL Correlations Dashboard: Across Season



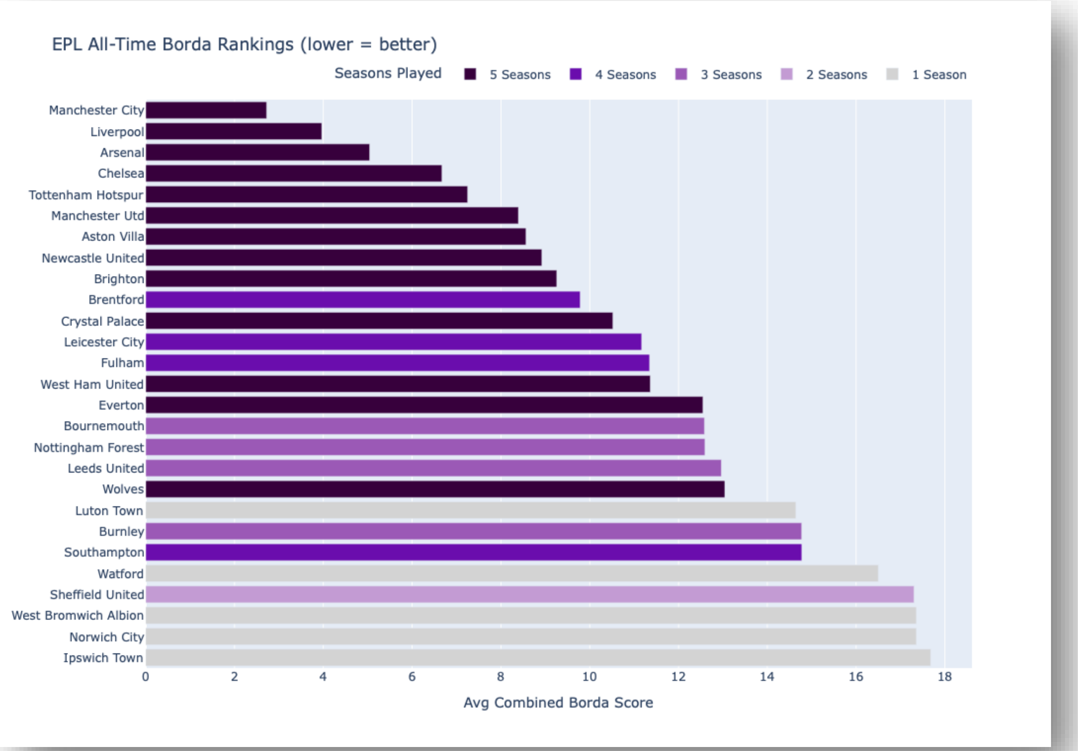


Seasons Reviewed: 2020—2024

Overview:

- Like our work with the NFL datasets, our group developed several interactive dashboards so that an end user could interface directly with the data to view how winning percentage and different offensive and defensive metrics correlate.
- The most glaring finding in the data was Manchester City's sustained dominance over the past five seasons. There is a significant gap between their statistical performance, consistency, and win percentage compared to even their main rivals.
- Liverpool has been able to achieve success; their season-to-season consistency has led to unfavorable seasons, as they have struggled to consistently produce statistical performance that more strongly correlates with winning.
- Finally, the data, as expected, reinforced the challenges that lower-tier teams face when promoted into the EPL. These teams were consistently lower across Borda scores and, as a result, were often relegated after just a single season in the EPL.
- Another significant finding in the discussion is that teams' offensive and defensive Borda scores have a strong negative correlation. Teams that perform well offensively tend to perform poorly on defense, and vice versa. Again, this warrants further research, but the financial stratification of the EPL likely plays a major role in these results.

EPL All-Time Borda Rankings (Avg Combined)



Seasons Reviewed: 2020—2024

Overview:

- The graph on the left is the most important graphic we made, as it really shows the difference between the four major professional sports leagues in the US and the EPL.
- The EPL has had 27 total teams compete over the past 5 seasons. Five teams have only been able to compete for a single season.
- This is where measuring season-to-season statistical performance consistency in the EPL differed greatly from that in the American leagues.
- All the teams with the highest consistency rankings have consistently competed in the EPL over the past five seasons. The main exception is Wolves, which has not been relegated but still has an average combined Borda metric in the lower half. This may indicate that the Wolves have structural challenges that may lead to an eventual relegation.

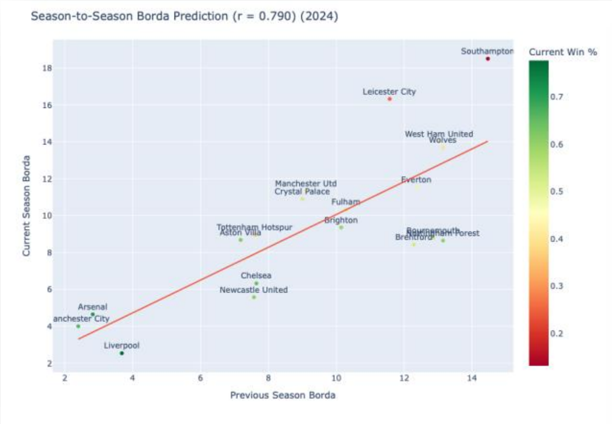
EPL Quality vs. Consistency

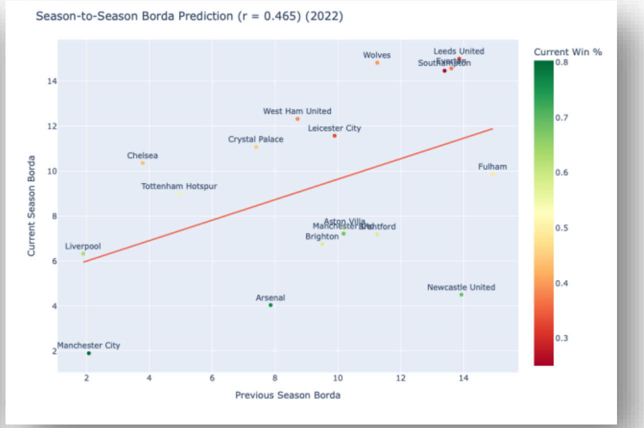
Seasons Reviewed: 2020—2024

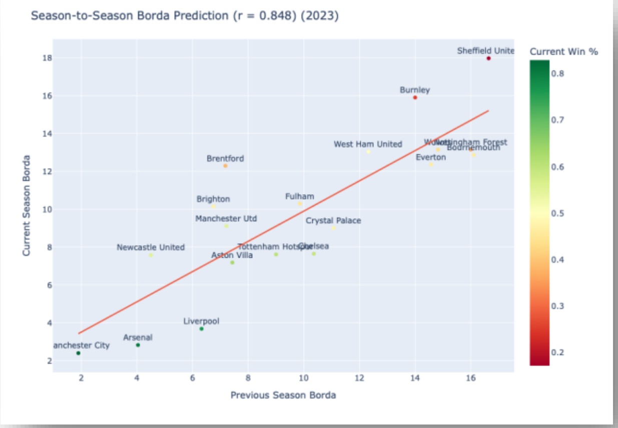
Overview:

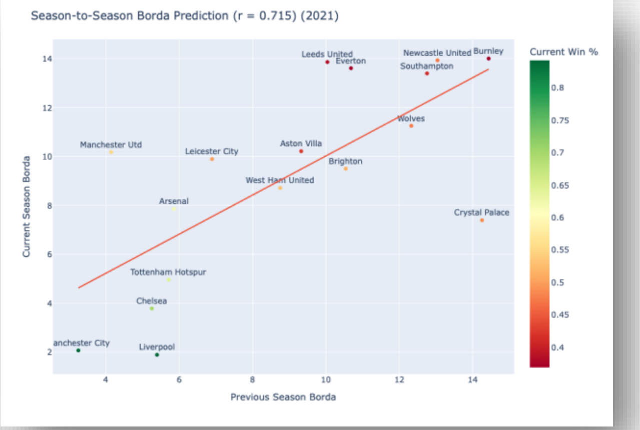
- The relationship between quality and consistency in the EPL differs significantly from that of its American counterparts. The EPL tends to be more predictable, with club success mainly driven by lower financial investment restrictions. The league most similar in how the financial strength of clubs correlates with statistical performance and win percentage is the MLB, which does not have a salary cap.
- This highlights that talent equality levels differ between the EPL and American sports leagues. Manchester City, along with Liverpool and Arsenal, are among the clubs spending the most globally. Nevertheless, Liverpool and Arsenal have faced more consistency challenges in their seasonal performance compared to Manchester City.
- Manchester United exemplifies well-funded clubs that lack a clear identity. Over the past five years, they have shown significant inconsistency, despite maintaining competitive Borda scores.

EPL Season-to-Season Correlations









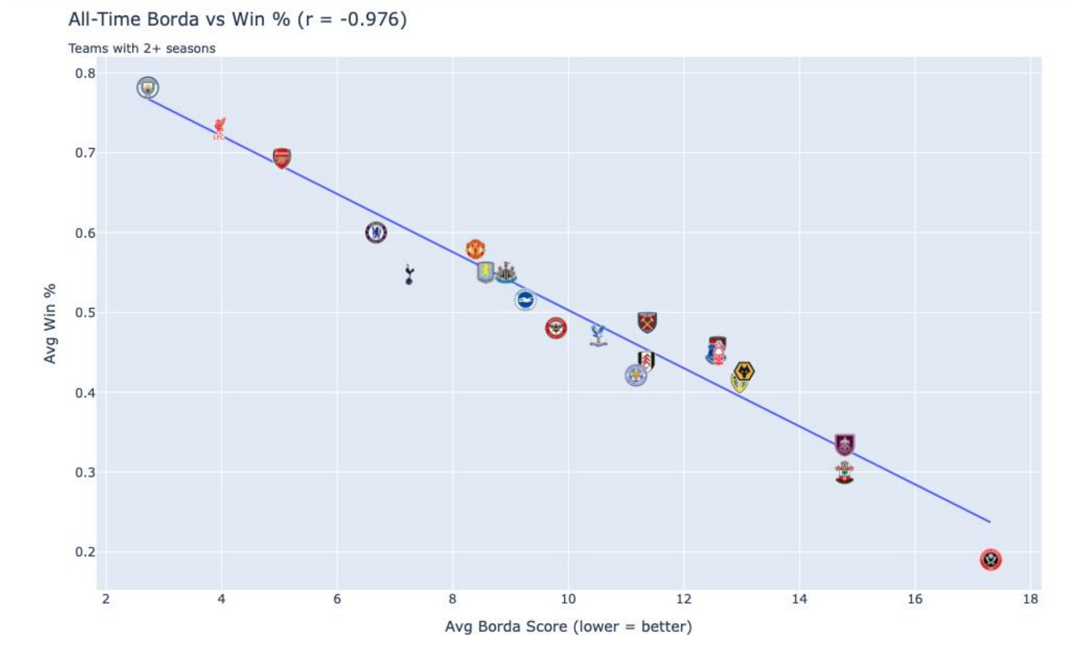
Seasons Reviewed: 2020—2024

Overview:

- The graphics on the left show the season-to-season Board score prediction (previous versus current season), which measures how well a team's previous-season Borda ranking correlates with its current-season statistical performance.
- As expected, there is a strong positive relationship between the previous-season Borda and current-season Borda results. The teams that are finishing at the top largely remain the same year-over-year. There is some variation between seasons, but overall, the top- and bottom-of-the-table teams are very predictable.
- The outlier season was the 2022 EPL season, in which r dropped by almost half compared to the 2021 season, then peaked in the 2023 season. This would warrant further investigations in future analyses.

EPL Borda Correlations: 2022

Avg Borda vs Win %	Season-to-Season Borda	Prev Borda → Current Win %	Teams Tracked
r = -0.979	r = +0.711	r = -0.661	27

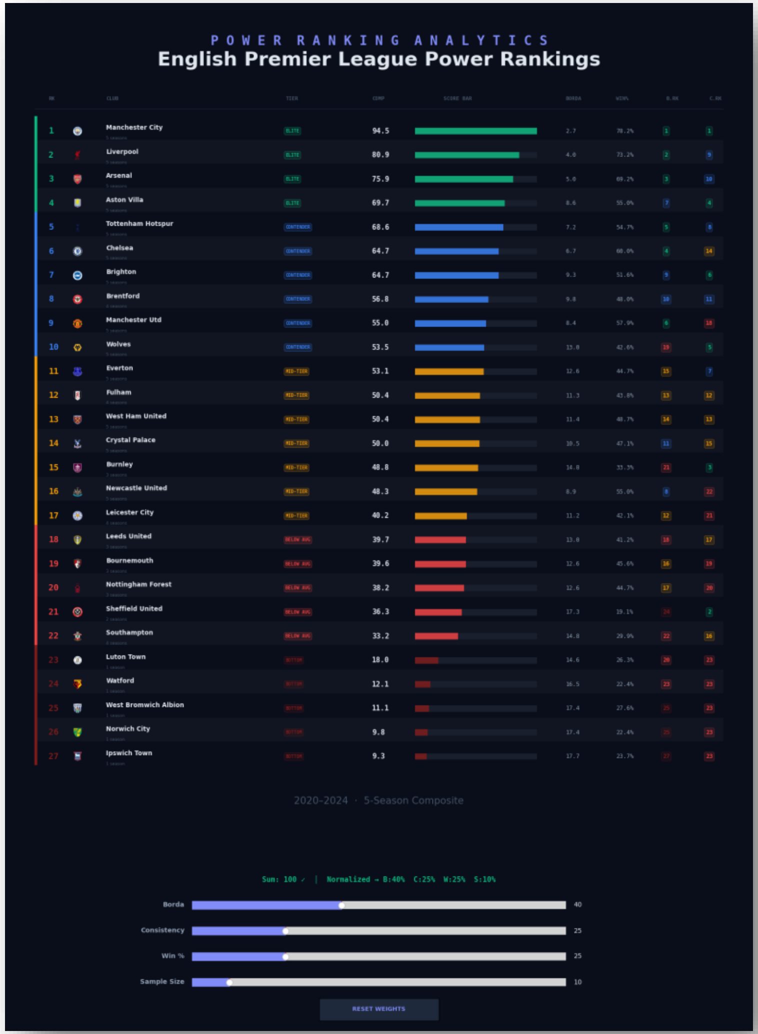


Seasons Reviewed: 2020—2024

Overview:

- These graphics on this page show how powerful the Borda ranking system can be for developing a power ranking of the EPL. Average Borda, Season-to-Season Borda, and Prev Borda to Current Win % all have strong correlations (regardless of direction).
- All-Borda vs Win Percentage is almost perfectly negatively correlated.
- However, it must be noted that this is a retrospective analysis where all information is known for each season. Just like in the other leagues, there would need to be some logic adjustments if this were used outside of developing unique and engaging talking points.

EPL Borda: Power Rankings



Seasons Reviewed: 2020—2024

Talking Points:

- **Manchester City's Dominance:** Before the 2024-2025 EPL season, Manchester City had won four consecutive EPL titles. This is not surprising given that their Borda Rank is #1 and their statistical performance has also consistently ranked #1. The only other team to win a championship during the dataset was Liverpool, who rank second in Borda. However, it is telling that Liverpool only ranked ninth in consistency.
- **Aston Villa – Consistent but lacks top end:** Aston Villa ranked 4th in composite scoring and landed in the elite category; however, that likely doesn't tell the club's full story. Villa is a club that ranked 4th in consistency but only 7th in Borda, demonstrating a team that is just below their rivals in their challenge for league supremacy. Until the club improves its Borda Scores, it will likely continue to finish in the top half of the table but never truly challenge for a title.
- **Promoted Clubs Struggle:** This is a long-known and unsurprising finding in the power rankings. Recently promoted teams struggled to stay up during their first season in the EPL. Based on the Borda scores, these teams just lack the quality to compete at the EPL level and don't have time to build a roster that is capable of consistently competing.
- **Champions are limited to top performers:** Unlike the NFL data previously seen, the EPL has much lower turnover among league winners. With only two teams, Manchester City and Liverpool, winning titles over the past five seasons.
- **Consistency doesn't equal success:** Just as seen with the NFL, teams can have very consistent statistical output, but that doesn't mean teams are consistently good. As is the case with Burnley, who over 3 seasons ranked 3rd in terms of consistency, but only 21st in Borda. Unfortunately, the conclusion is that when Burnley reach the EPL, they consistently perform poorly.

NBA OVERVIEW

All Data Gathered from [Pro-Basketball Reference]

NBA Python Code: CSVs & Data Wrangling

```python
per_game = pd.read_csv("nba_team_per_game_2021_2025.csv")
advanced = pd.read_csv("nba_team_advanced_2021_2025.csv")
nba = per_game.merge(advanced, on=["Team","Season"])
nba["W_pct"] = nba["W"] / (nba["W"] + nba["L"])
```

```python
nba.head()
nba.info()
nba.describe()
```

	Season	Team	G	MP	FG	FGA	FG%	3P	3PA	3P%	...	3PAr	TS%	eFG%	TOV%	ORB%	FT/FGA	Arena	Attend.	Attend./G	W_pct
0	2024-25	Atlanta Hawks	82.0	241.2	43.4	91.8	0.472	13.5	37.7	0.358	...	0.410	0.579	0.546	13.2	26.3	0.196	State Farm Arena	657613.0	16440.0	0.487805
1	2024-25	Boston Celtics	82.0	241.8	41.6	90.0	0.462	17.8	48.2	0.368	...	0.536	0.591	0.561	10.8	25.7	0.169	TD Garden	785396.0	19156.0	0.743902
2	2024-25	Brooklyn Nets	82.0	241.2	37.6	86.1	0.437	13.6	39.4	0.344	...	0.458	0.552	0.516	13.7	24.5	0.189	Barclays Center	713304.0	17398.0	0.317073
3	2024-25	Charlotte Hornets	82.0	240.9	38.3	89.1	0.430	13.0	38.3	0.339	...	0.430	0.537	0.503	13.7	26.5	0.174	Spectrum Center	703935.0	17169.0	0.231707
4	2024-25	Chicago Bulls	82.0	240.9	43.2	92.0	0.470	15.4	42.0	0.367	...	0.457	0.585	0.553	12.7	22.3	0.173	United Center	825659.0	20138.0	0.475610

5 rows × 48 columns

```python
target = "W_pct"
nba_num = nba.select_dtypes(include=[np.number])
drop_like = nba_num.columns[nba_num.columns.str.match(r"^(W($|_)|L($|_))")].tolist()
predictors = nba_num.drop(columns=list(set(drop_like + [target])), errors="ignore")
non_offense_cols = [
    "NRtg",
    "ORtg",
    "DRtg",
    "MOV",
    "SRS",
    "Age",
    "PW",
    "PL",
    "MP",
    "SOS",
    "Attend.", "Attend./G"]
predictors = predictors.drop(columns=non_offense_cols, errors="ignore")
y = nba_num[target]
predictors.shape
```

```
(150, 30)
```

```python
corr_pred = pd.concat([predictors, nba_num[[target]]], axis=1).corr()
win_corr = corr_pred[target].drop(target).sort_values(ascending=False)
win_corr.head(20)
```

```
TS%       0.705123
eFG%      0.700162
PTS       0.598510
DRB       0.343608
3P        0.259693
TRB       0.216125
AST       0.205588
FG        0.186016
STL       0.155927
FT/FGA    0.121140
3PAr      0.119635
FT        0.105921
3PA       0.096699
2P        0.088303
BLK       0.065557
ORB%      0.055209
FTA       0.008096
FGA       0.006813
FTr       0.004009
FG%      -0.002838
Name: W_pct, dtype: float64
```

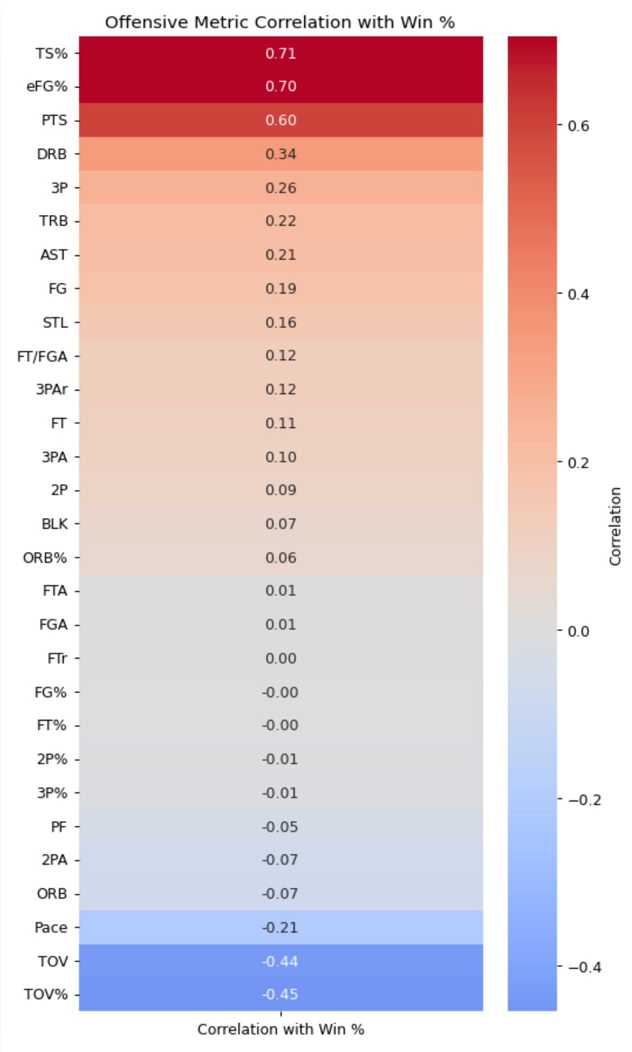
NBA Python Code: Heatmaps & Correlations

```
exclude_cols = ["G", "MP"]
win_corr_clean = win_corr.drop(labels=exclude_cols, errors="ignore")
top15 = win_corr_clean.sort_values(ascending=False).head(15)
bottom15_raw = win_corr_clean.sort_values(ascending=True).head(14)
bottom15 = bottom15_raw.sort_values(ascending=False)
ordered_stats = list(top15.index) + list(bottom15.index)
win_corr_30 = win_corr_clean.loc[ordered_stats].to_frame(name="Correlation with Win %")
plt.figure(figsize=(6,10))
sns.heatmap(
    win_corr_30,
    cmap="coolwarm",
    center=0,
    annot=True,
    fmt=".2f",
    cbar_kws={"label": "Correlation"})
plt.title("Offensive Metric Correlation with Win %")
plt.ylabel("")
plt.tight_layout()
plt.show()
```

```
bottom_corr = win_corr.drop("G").tail(10)
plt.figure(figsize=(10,8))
bottom_corr.sort_values().plot(kind="barh")
plt.title("Bottom 10 Offensive Stats Correlated with Win %")
plt.xlabel("Correlation with Win %")
plt.ylabel("Offensive Stat")
plt.tight_layout()
plt.show()
```

```
top_corr = win_corr.head(10)
plt.figure(figsize=(10,8))
top_corr.sort_values().plot(kind="barh")
plt.title("Top 10 Offensive Stats Correlated with Win %")
plt.xlabel("Correlation with Win %")
plt.ylabel("Offensive Stat")
plt.tight_layout()
plt.show()
```

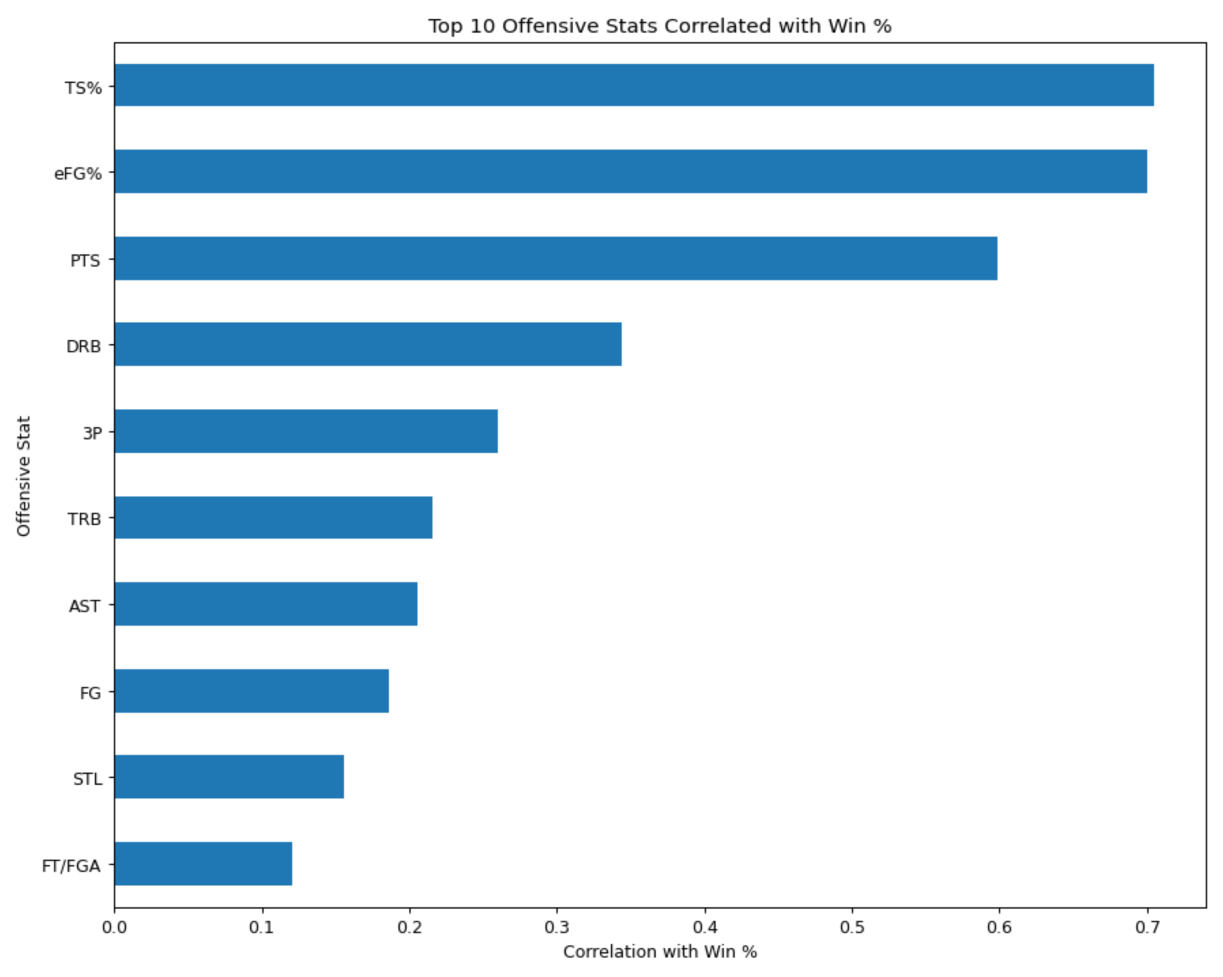
NBA Offensive Correlations: Heatmap



Seasons Reviewed: 2020—2025

Findings: To begin the project, I wanted to understand which offensive statistics are actually most associated with winning in the NBA. To do that, I calculated the Pearson correlation coefficient between each offensive team statistic and team win percentage across the last five complete NBA seasons. The Pearson correlation measures the linear relationship between two variables, where positive values indicate that as a statistic increases, win percentage tends to increase, and where negative values indicate that as a statistic increases, win percentage tends to decrease. After computing these correlations for every offensive metric in my dataset, I visualized them in this heatmap, where red indicates stronger positive correlations with winning, and blue indicates negative relationships.

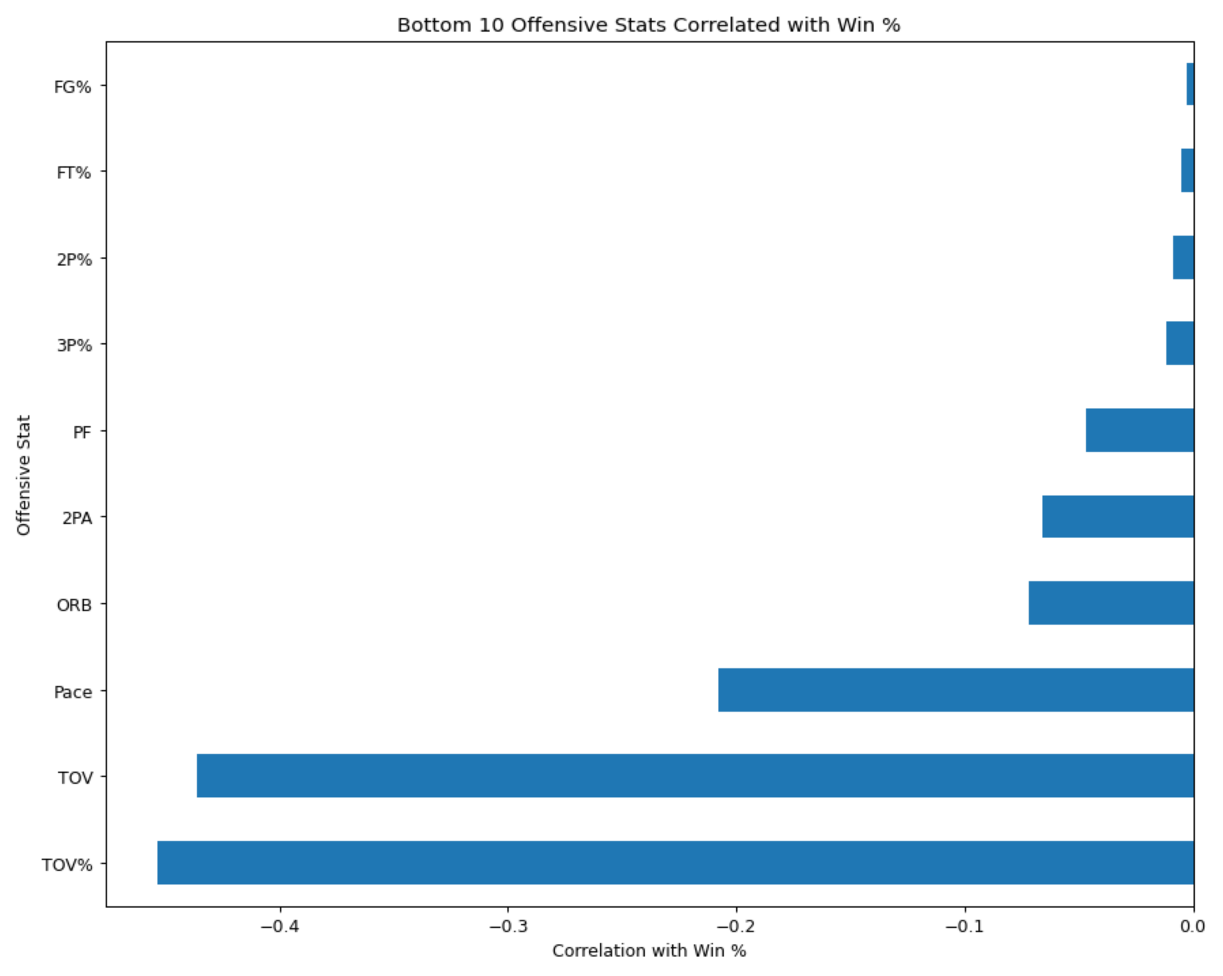
NBA Offensive Positive Correlations



Seasons Reviewed: 2020—2025

Overview: Both TS% and eFG% measure scoring efficiency, rather than just raw scoring volume, where TS% incorporates free throws, two point, and three point shots into one efficiency metric, while eFG% adjusts traditional field goal percentage to account for the added value of three point shots. These results makes intuitive sense because efficient scoring is one of the strongest indicators of offensive success in modern basketball. Points per game also appears highly correlated because teams that consistently score more points obviously tend to win more games, but it’s slightly less predictive than efficiency metrics.

NBA Offensive Negative Correlations



Seasons Reviewed: 2020—2025

Overview: Looking at the bottom three metrics we see TOV%, total turnovers, and pace. The strong negative correlation for turnovers makes sense because turnovers represent lost possessions, which directly reduces scoring opportunities while often creating transition opportunities for the opponent. Pace is interesting because it measures the number of possessions per game rather than efficiency. Faster teams are not necessarily better teams, and in many cases weaker teams play faster to increase variance or compensate for lower half-court efficiency, which likely explains the slightly negative relationship we see here.

NBA Python Code: Borda Power Ranking

```python
per_path = "nba_team_per_game_2021_2025.csv"
adv_path = "nba_team_advanced_2021_2025.csv"
per = pd.read_csv(per_path)
adv = pd.read_csv(adv_path)
nba = per.merge(adv, on=["Season", "Team"], how="inner")
nba["W_pct"] = nba["W"] / (nba["W"] + nba["L"])
OFF_METRICS = {
    "ORtg": True,
    "TS%": True,
    "eFG%": True,
    "ORB%": True,
    "FT/FGA": True,
    "3PAr": True,
    "TOV%": False}
DEF_METRICS = {
    "DRtg": False}
def rank_metric(series: pd.Series, higher_better: bool) -> pd.Series:
    return series.rank(ascending=not higher_better, method="min")
season_df = nba.copy()
for col, higher_better in OFF_METRICS.items():
    season_df[f"rk_{col}"] = season_df.groupby("Season")[col].transform(
        lambda s: rank_metric(s, higher_better))
for col, higher_better in DEF_METRICS.items():
    season_df[f"rk_{col}"] = season_df.groupby("Season")[col].transform(
        lambda s: rank_metric(s, higher_better))
season_df["Off_Borda_Score"] = season_df[[f"rk_{c}" for c in OFF_METRICS]].mean(axis=1)
season_df["Def_Borda_Score"] = season_df[[f"rk_{c}" for c in DEF_METRICS]].mean(axis=1)
season_df["Combined_Borda_Score"] = season_df[["Off_Borda_Score", "Def_Borda_Score"]].mean(axis=1)
nba_cross = season_df.groupby("Team").agg(
    Seasons_Played=("Season", "nunique"),
    Avg_W_pct=("W_pct", "mean"),
    Avg_Combined_Borda=("Combined_Borda_Score", "mean"),
    Combined_Borda_StdDev=("Combined_Borda_Score", "std"),).reset_index()
nba_cross["Avg_W_pct"] = nba_cross["Avg_W_pct"] * 100
nba_cross["Combined_Borda_StdDev"] = nba_cross["Combined_Borda_StdDev"].fillna(0)
nba_cross["Overall_Combined_Rank"] = nba_cross["Avg_Combined_Borda"].rank(
    ascending=True, method="min").astype(int)
nba_cross["Consistency_Rank"] = nba_cross["Combined_Borda_StdDev"].replace(0, np.nan).rank(
    ascending=True, method="min", na_option="bottom").astype(int)
df = nba_cross.rename(columns={"Team": "Tm"})
df.head()
```

```python
import matplotlib.pyplot as plt
from matplotlib.widgets import Slider, Button
from matplotlib.offsetbox import OffsetImage, AnnotationBbox
from PIL import Image
import requests
from io import BytesIO
import numpy as np
n_teams = len(df)
min_borda = df['Avg_Combined_Borda'].min()
max_borda = df['Avg_Combined_Borda'].max()
max_consist = df['Consistency_Rank'].max()
abbr_map = {
    "Atlanta Hawks": "atl",
    "Boston Celtics": "bos",
    "Brooklyn Nets": "bkn",
    "Charlotte Hornets": "cha",
    "Chicago Bulls": "chi",
    "Cleveland Cavaliers": "cle",
    "Dallas Mavericks": "dal",
    "Denver Nuggets": "den",
    "Detroit Pistons": "det",
    "Golden State Warriors": "gs",
    "Houston Rockets": "hou",
    "Indiana Pacers": "ind",
    "Los Angeles Clippers": "lac",
    "Los Angeles Lakers": "lal",
    "Memphis Grizzlies": "mem",
    "Miami Heat": "mia",
    "Milwaukee Bucks": "mil",
    "Minnesota Timberwolves": "min",
    "New Orleans Pelicans": "no",
    "New York Knicks": "ny",
    "Oklahoma City Thunder": "okc",
    "Orlando Magic": "orl",
    "Philadelphia 76ers": "phi",
    "Phoenix Suns": "phx",
    "Portland Trail Blazers": "por",
    "Sacramento Kings": "sac",
    "San Antonio Spurs": "sa",
    "Toronto Raptors": "tor",
    "Utah Jazz": "utah",
    "Washington Wizards": "wsh",}
```

```python
missing = sorted(set(df["Tm"]) - set(abbr_map.keys()))
if missing:
    print("Teams in df missing from abbr_map:", missing)
team_colors = {
    "atl": "#E03A3E",
    "bos": "#007A33",
    "bkn": "#000000",
    "cha": "#1D1160",
    "chi": "#CE1141",
    "cle": "#6F263D",
    "dal": "#00538C",
    "den": "#0E2240",
    "det": "#C8102E",
    "gs":  "#1D428A",
    "hou": "#CE1141",
    "ind": "#002D62",
    "lac": "#C8102E",
    "lal": "#552583",
    "mem": "#5D76A9",
    "mia": "#98002E",
    "mil": "#00471B",
    "min": "#0C2340",
    "no":  "#0C2340",
    "ny":  "#006BB6",
    "okc": "#007AC1",
    "orl": "#0077C0",
    "phi": "#006BB6",
    "phx": "#1D1160",
    "por": "#E03A3E",
    "sac": "#5A2D81",
    "sa":  "#C4CED4",
    "tor": "#CE1141",
    "utah":"#002B5C",
    "wsh":"#002B5C",}
def fetch_logo(abbr, size=28):
    try:
        url = f"https://a.espncdn.com/combiner/i?img=/i/teamlogos/nba/500/{abbr}.png&h=80&w=80"
        resp = requests.get(url, timeout=5)
        img = Image.open(BytesIO(resp.content)).convert("RGBA").resize((size, size), Image.LANCZOS)
        return np.array(img)
    except:
        return None
print("Fetching team logos and loading visualization...")
logos = {abbr: fetch_logo(abbr) for abbr in abbr_map.values()}
```

NBA Python Code: Borda Power Ranking

```python
def compute_composite(row, weights, season_penalty=True):
    norm_borda = 1 - (row['Avg_Combined_Borda'] - min_borda) / (max_borda - min_borda)
    norm_consist = 1 - (row['Consistency_Rank'] - 1) / (max_consist - 1)
    norm_wpct = row['Avg_W_pct'] / 100
    season_factor = min(row['Seasons_Played'] / 3, 1.0) if season_penalty else 1.0
    raw = (norm_borda * weights['borda'] +
           norm_consist * weights['consistency'] +
           norm_wpct * weights['win_pct'])
    return round(raw, 5)
def get_tier(rank, total):
    if rank <= 5:  return ('ELITE', '#10b981')
    if rank <= 12: return ('CONTENDER', '#3b82f6')
    if rank <= 20: return ('MID-TIER', '#f59e0b')
    if rank <= 28: return ('BELOW AVG', '#ef4444')
    return ('BOTTOM', '#7f1d1d')
fig = plt.figure(figsize=(16, 24))
fig.patch.set_facecolor('#0a0e1a')
ax = fig.add_axes([0.02, 0.11, 0.96, 0.82])
ax.set_facecolor('#0a0e1a')
ax.axis('off')
slider_color = '#1e293b'
active_color = '#818cf8'
ax_borda   = fig.add_axes([0.25, 0.095, 0.50, 0.015], facecolor=slider_color)
ax_consist = fig.add_axes([0.25, 0.070, 0.50, 0.015], facecolor=slider_color)
ax_wpct    = fig.add_axes([0.25, 0.045, 0.50, 0.015], facecolor=slider_color)
s_borda   = Slider(ax_borda,   'Borda',       0, 100, valinit=45, valstep=1, color=active_color)
s_consist = Slider(ax_consist, 'Consistency', 0, 100, valinit=30, valstep=1, color=active_color)
s_wpct    = Slider(ax_wpct,    'Win %',       0, 100, valinit=25, valstep=1, color=active_color)
for s in [s_borda, s_consist, s_wpct]:
    s.label.set_color('#94a3b8')
    s.label.set_fontsize(10)
    s.label.set_fontweight('bold')
    s.valtext.set_color('#e2e8f0')
    s.valtext.set_fontsize(10)
weight_text = fig.text(0.50, 0.120, '', ha='center', fontsize=11,
                       fontweight='bold', fontfamily='monospace')
```

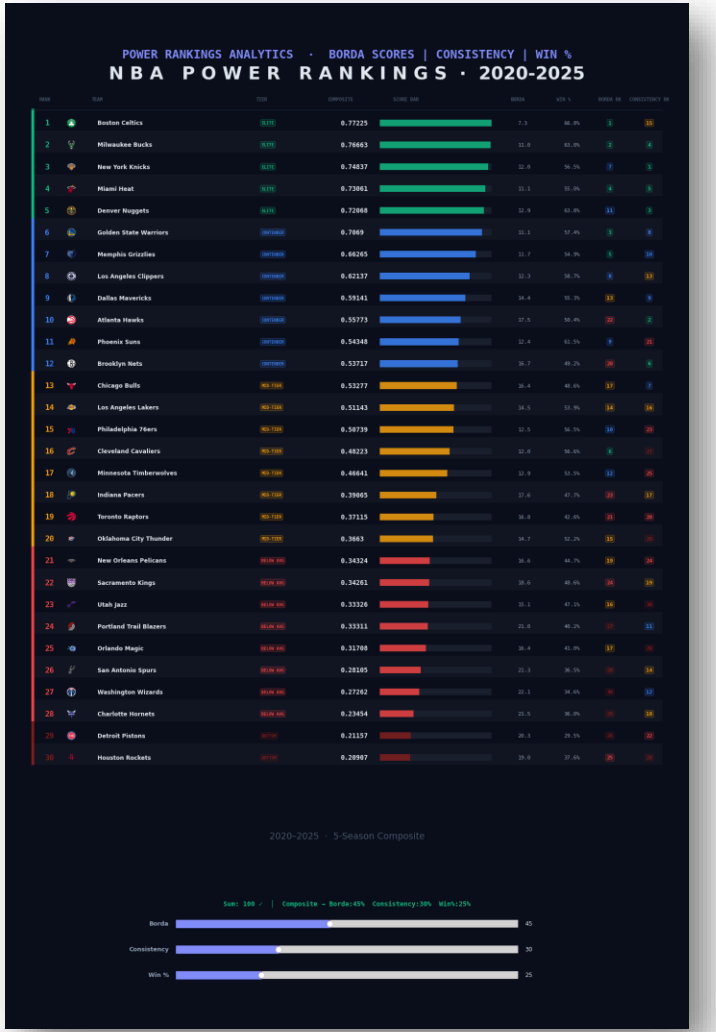
```python
def update(val=None):
    ax.clear()
    ax.set_facecolor('#0a0e1a')
    ax.axis('off')
    ax.set_xlim(0, 1)
    ax.set_ylim(0, 1)
    raw_b, raw_c, raw_w = s_borda.val, s_consist.val, s_wpct.val
    total = raw_b + raw_c + raw_w
    if total == 0:
        weights = {'borda': 0.25, 'consistency': 0.25, 'win_pct': 0.25}
        total = 100
    else:
        weights = {
            'borda': raw_b / total,
            'consistency': raw_c / total,
            'win_pct': raw_w / total,}
    warn = '!' if abs(total - 100) > 0.5 else '√'
    weight_text.set_text(
        f"Sum: {int(total)} {warn}  |  "
        f"Composite → Borda:{weights['borda']:.0%}  Consistency:{weights['consistency']:.0%}  "
        f"Win%:{weights['win_pct']:.0%}")
    weight_text.set_color('#10b981' if abs(total - 100) < 1 else '#f59e0b')
    df['Composite'] = df.apply(compute_composite, axis=1, weights=weights)
    ranked = df.sort_values('Composite', ascending=False).reset_index(drop=True)
    ranked['Power_Rank'] = range(1, n_teams + 1)
    max_comp = ranked['Composite'].max()
    row_h = 0.026
    header_y = 0.97
    start_y = header_y - 0.028
    ax.text(0.5, 1.03, 'POWER RANKINGS ANALYTICS  ·  BORDA SCORES | CONSISTENCY | WIN %', transform=ax.transAxes,
            ha='center', va='top', fontsize=20, fontweight='bold', color='#818cf8',
            fontfamily='monospace')
    ax.text(0.5, 1.01, 'N B A   P O W E R   R A N K I N G S  ·  2020-2025', transform=ax.transAxes,
            ha='center', va='top', fontsize=30, fontweight='bold', color='#e2e8f0')
    ax.text(0.5, 0.100, '2020-2025  ·  5-Season Composite',
            transform=ax.transAxes, ha='center', va='top', fontsize=15, color='#475569')
    cols = {'rank': 0.04, 'logo': 0.08, 'team': 0.12, 'tier': 0.37,
            'composite': 0.49, 'bar_start': 0.55, 'bar_end': 0.72,
            'borda': 0.76, 'wpct': 0.83, 'borda_rk': 0.90, 'consist_rk': 0.96}
    for x, label in [(cols['rank'], 'RANK'), (cols['team'], 'TEAM'), (cols['tier'], 'TIER'),
                     (cols['composite'], 'COMPOSITE'), (cols['bar_start'] + 0.04, 'SCORE BAR'),
                     (cols['borda'], 'BORDA'), (cols['wpct'], 'WIN %'),
                     (cols['borda_rk'], 'BORDA RK'), (cols['consist_rk'], 'CONSISTENCY RK')]:
        ax.text(x, header_y, label, ha='center', transform=ax.transAxes, fontsize=8,
                fontweight='bold', color='#475569', fontfamily='monospace', va='center')
    ax.plot([0.02, 0.98], [header_y - 0.012, header_y - 0.012],
            transform=ax.transAxes, color='#1e293b', linewidth=0.8)
```

```python
    for i, (_, row) in enumerate(ranked.iterrows()):
        y = start_y - i * row_h
        tm = row['Tm']
        abbr = abbr_map[tm]
        tier_label, tier_color = get_tier(row['Power_Rank'], n_teams)
        if i % 2 == 1:
            r = plt.Rectangle((0.02, y - row_h/2 + 0.004), 0.96, row_h,
                              transform=ax.transAxes, facecolor='#ffffff08', edgecolor='none', zorder=0)
            ax.add_patch(r)
        a = plt.Rectangle((0.02, y - row_h/2 + 0.004), 0.004, row_h,
                          transform=ax.transAxes, facecolor=tier_color, edgecolor='none', zorder=1)
        ax.add_patch(a)
        ax.text(cols['rank'], y, str(row['Power_Rank']), transform=ax.transAxes,
                fontsize=13, fontweight='bold', color=tier_color, va='center', fontfamily='monospace')
        logo_arr = logos.get(abbr)
        if logo_arr is not None:
            im = OffsetImage(logo_arr, zoom=0.55)
            ab = AnnotationBbox(im, (cols['logo'], y), xycoords='axes fraction',
                                frameon=False, box_alignment=(0.5, 0.5))
            ax.add_artist(ab)
        ax.text(cols['team'], y, tm, transform=ax.transAxes,
                fontsize=9.5, fontweight='bold', color='#e2e8f0', va='center')
        bp = dict(boxstyle='round,pad=0.3', facecolor=tier_color + '22', edgecolor=tier_color + '66', linewidth=0.8)
        ax.text(cols['tier'], y, tier_label, transform=ax.transAxes, fontsize=7,
                fontweight='bold', color=tier_color, va='center', bbox=bp, fontfamily='monospace')
        ax.text(cols['composite'], y, f"{row['Composite']}", transform=ax.transAxes,
                fontsize=11, fontweight='bold', color='#e2e8f0', va='center', fontfamily='monospace')
        bw = (cols['bar_end'] - cols['bar_start']) * (row['Composite'] / max_comp) if max_comp > 0 else 0
        bg = plt.Rectangle((cols['bar_start'], y - 0.004), cols['bar_end'] - cols['bar_start'],
                          0.008, transform=ax.transAxes, facecolor='#1a1f2e', edgecolor='none', zorder=1)
        fl = plt.Rectangle((cols['bar_start'], y - 0.004), bw, 0.008,
                          transform=ax.transAxes, facecolor=tier_color, edgecolor='none', zorder=2, alpha=0.85)
        ax.add_patch(bg)
        ax.add_patch(fl)
        ax.text(cols['borda'], y, f"{row['Avg_Combined_Borda']:.1f}", transform=ax.transAxes,
                fontsize=9, color='#94a3b8', va='center', fontfamily='monospace')
        ax.text(cols['wpct'], y, f"{row['Avg_W_pct']:.1f}%", transform=ax.transAxes,
                fontsize=9, color='#94a3b8', va='center', fontfamily='monospace')
        for cx, v in [(cols['borda_rk'], row['Overall_Combined_Rank']),
                      (cols['consist_rk'], row['Consistency_Rank'])]:
            p = (n_teams - v) / (n_teams - 1)
            bc = '#10b981' if p > 0.8 else '#3b82f6' if p > 0.6 else '#f59e0b' if p > 0.35 else '#ef4444' if p > 0.15 else '#7f1d1d'
            bbp = dict(boxstyle='round,pad=0.25', facecolor=bc + '22', edgecolor=bc + '55', linewidth=0.7)
            ax.text(cx, y, str(int(v)), transform=ax.transAxes, fontsize=9,
                    fontweight='bold', color=bc, va='center', ha='center', bbox=bbp, fontfamily='monospace')
    fig.canvas.draw_idle()
s_borda.on_changed(update)
s_consist.on_changed(update)
s_wpct.on_changed(update)
update()
plt.show()
```

```
Fetching team logos and loading visualization...
```

Borda score represents a team's overall statistical performance across multiple metrics, and teams were ranked within each season for every selected statistic relative to the rest of the league. Those ranks were then averaged together to create an offensive Borda score, and defensive ratings were incorporated to form a combined Borda score. Because the Borda score is based on average ranking across many metrics, a lower score indicates stronger overall performance. The consistency rank measures how stable a team's performance has been overtime, and was calculated using the standard deviation of each team's combined Borda score across seasons. A low standard deviation means a team performed at a similar level every year, indicating consistent performance, and a high standard deviation means the team's performance fluctuates significantly from season to season. Teams were then ranked based on this variability, with Rank 1 representing the most statistically consistent team across the five year period. Lastly, the final component is average win percentage, which was included to ensure that the model still accounts for actual game outcomes, but overall we wanted the broader statistical profile to play a larger role in determining overall team strength which is why we felt these were the right weights for each component.

NBA Borda Power Rankings



Seasons Reviewed: 2020—2025

Overview: After identifying which statistics are most associated with winning, I used those insights to build a multi-metric power ranking system. At the very bottom you can see that this model combines three components: Borda score, consistency rank, and average win percentage. These metrics all compute one composite score which is used to rank the teams 1 through 30. The teams are grouped into performance tiers based on their composite scores. At the top of the rankings we see teams like the Celtics, Bucks, Knicks, Heat, and Nuggets. These teams consistently rank highly because they combine strong efficiency metrics, positive net ratings, solid defensive performance, and relatively stable results across seasons. At the other end of the rankings, we see teams like the Rockets, Pistons, and Hornets. These teams rank lower primarily because of poor efficiency metrics, higher turnover rates, weaker defensive ratings, and inconsistent performance across the five year period. Since the model evaluates performance across multiple dimensions, teams that struggle in several statistical categories tend to fall toward the bottom.

MLB OVERVIEW

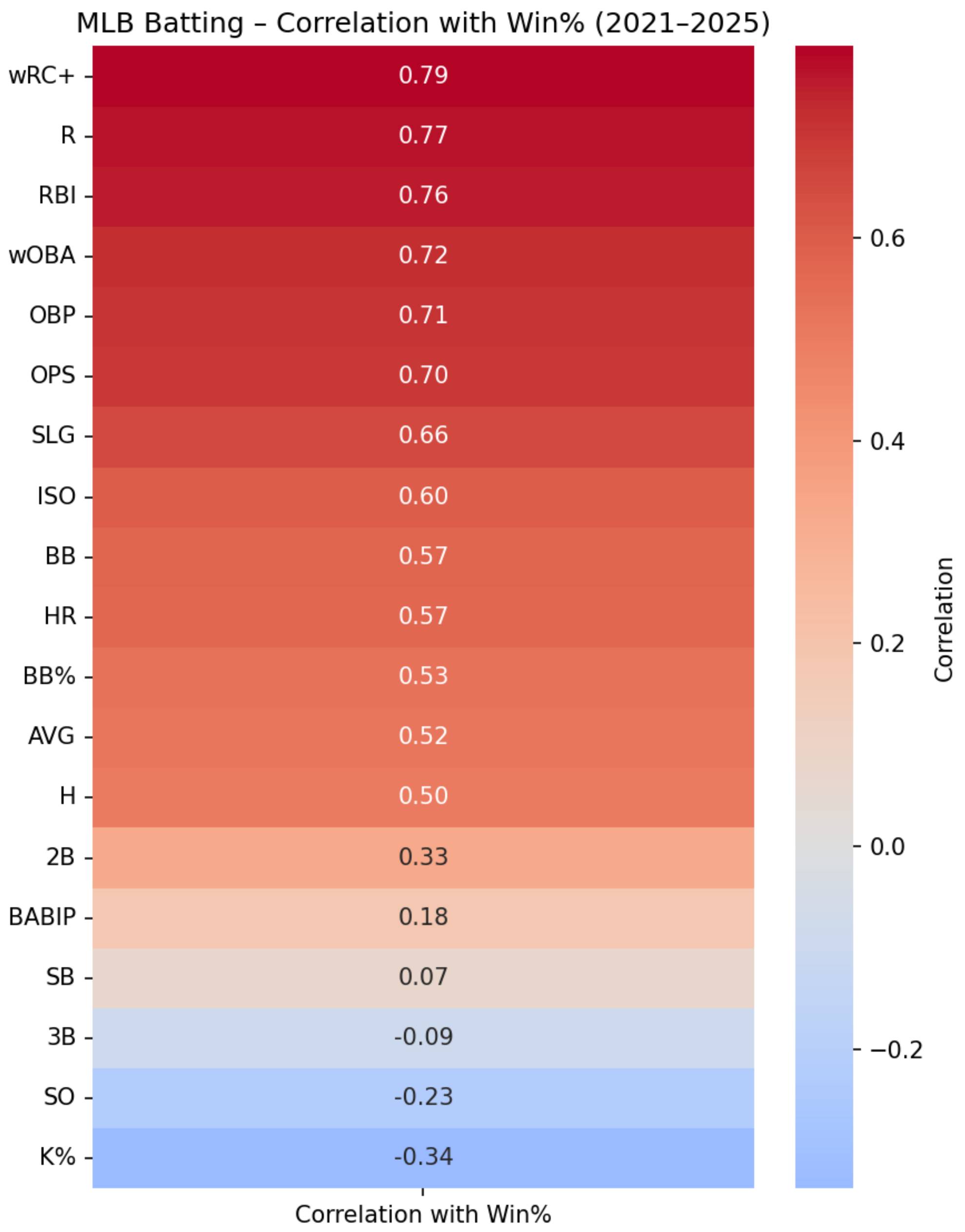
All Data Gathered from FanGraphs and Baseball-Reference (2021-2025 MLB seasons)

MLB Code

The data was imported from FanGraphs and Baseball-Reference using pandas and pybaseball. I utilized various other libraries to produce the graphs and images displayed in subsequent slides.

```
# —— Hardcoded W/L records 2021–2025 (Baseball-Reference final standings) ————
import pandas as pd

wl_data = [
    (2021,'Arizona Diamondbacks',52,110),(2021,'Atlanta Braves',88,73),
    (2021,'Baltimore Orioles',52,110),(2021,'Boston Red Sox',92,70),
    (2021,'Chicago Cubs',71,91),(2021,'Chicago White Sox',93,69),
    (2021,'Cincinnati Reds',83,79),(2021,'Cleveland Guardians',80,82),
    (2021,'Colorado Rockies',74,87),(2021,'Detroit Tigers',77,85),
    (2021,'Houston Astros',95,67),(2021,'Kansas City Royals',74,88),
    (2021,'Los Angeles Angels',77,85),(2021,'Los Angeles Dodgers',106,56),
    (2021,'Miami Marlins',67,95),(2021,'Milwaukee Brewers',95,67),
    (2021,'Minnesota Twins',73,89),(2021,'New York Mets',77,85),
    (2021,'New York Yankees',92,70),(2021,'Oakland Athletics',86,76),
    (2021,'Philadelphia Phillies',82,80),(2021,'Pittsburgh Pirates',61,101),
    (2021,'San Diego Padres',79,83),(2021,'San Francisco Giants',107,55),
    (2021,'Seattle Mariners',90,72),(2021,'St. Louis Cardinals',90,72),
    (2021,'Tampa Bay Rays',100,62),(2021,'Texas Rangers',60,102),
    (2021,'Toronto Blue Jays',91,71),(2021,'Washington Nationals',65,97),
```

```
# —— Pull FanGraphs team batting, pitching & baserunning via pybaseball ————
from pybaseball import team_batting, team_pitching, team_batting_bref, cache
cache.enable()

batting  = team_batting(2021, 2025)
pitching = team_pitching(2021, 2025)

# team_batting_bref has BsR, XBT%, SB, CS — pull season by season
bsr_frames = []
for yr in range(2021, 2026):
    try:
        df_yr = team_batting_bref(yr, yr)
        df_yr['Season'] = yr
        bsr_frames.append(df_yr)
```

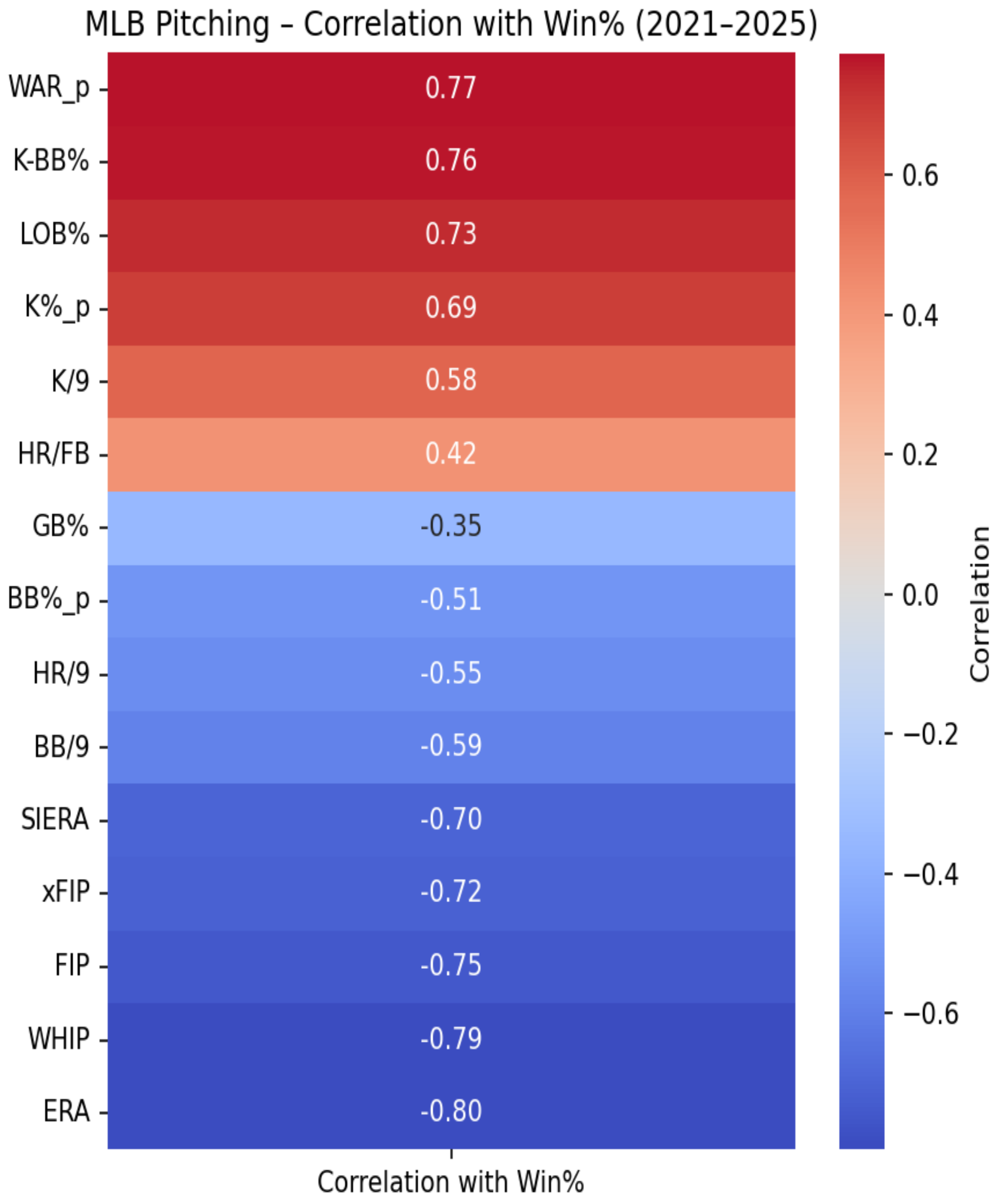


	NAME	wRC+	SEASON	TM	G	AB
1	Barry Bonds	244	2002	SFG	143	403
2	Babe Ruth	237	1920	NYY	142	458
3	Barry Bonds	234	2001	SFG	153	476
4	Barry Bonds	233	2004	SFG	147	373
5	Babe Ruth	230	1923	NYY	152	522
6	Babe Ruth	224	1921	NYY	152	540
7	Ted Williams	223	1957	BOS	132	420
8	Ted Williams	221	1941	BSA	143	456
9	Rogers Hornsby	221	1924	SLN	143	536
10	Aaron Judge	219	2024	NYY	158	559

Run production yields a high correlation with winning and strikeouts correlate with an increased odds of losing.

WRC+ is a solid indicator of offensive success. Both Bonds (2002) and Judge (2024) led teams to the world series each respective season.

HR's only yield a 57% correlation with winning despite a –34% of losing associated with an increased strikeout rate.



Team pitching staffs that consistently post strong strikeout to walk ratios typically rank amongst the top teams in the league standings.

FIP and WHIP remain excellent indicators of which pitchers to avoid given an increase in each metric typically results in a strong correlation with losing. It seems pitching roster decisions have the potential to be fueled by data and performance avoidance rather than data inspired acquisition.

Perhaps the most significant result in the MLB analysis was that a such an extreme 80% negative correlation existed when teams allowed an increase in ERA.

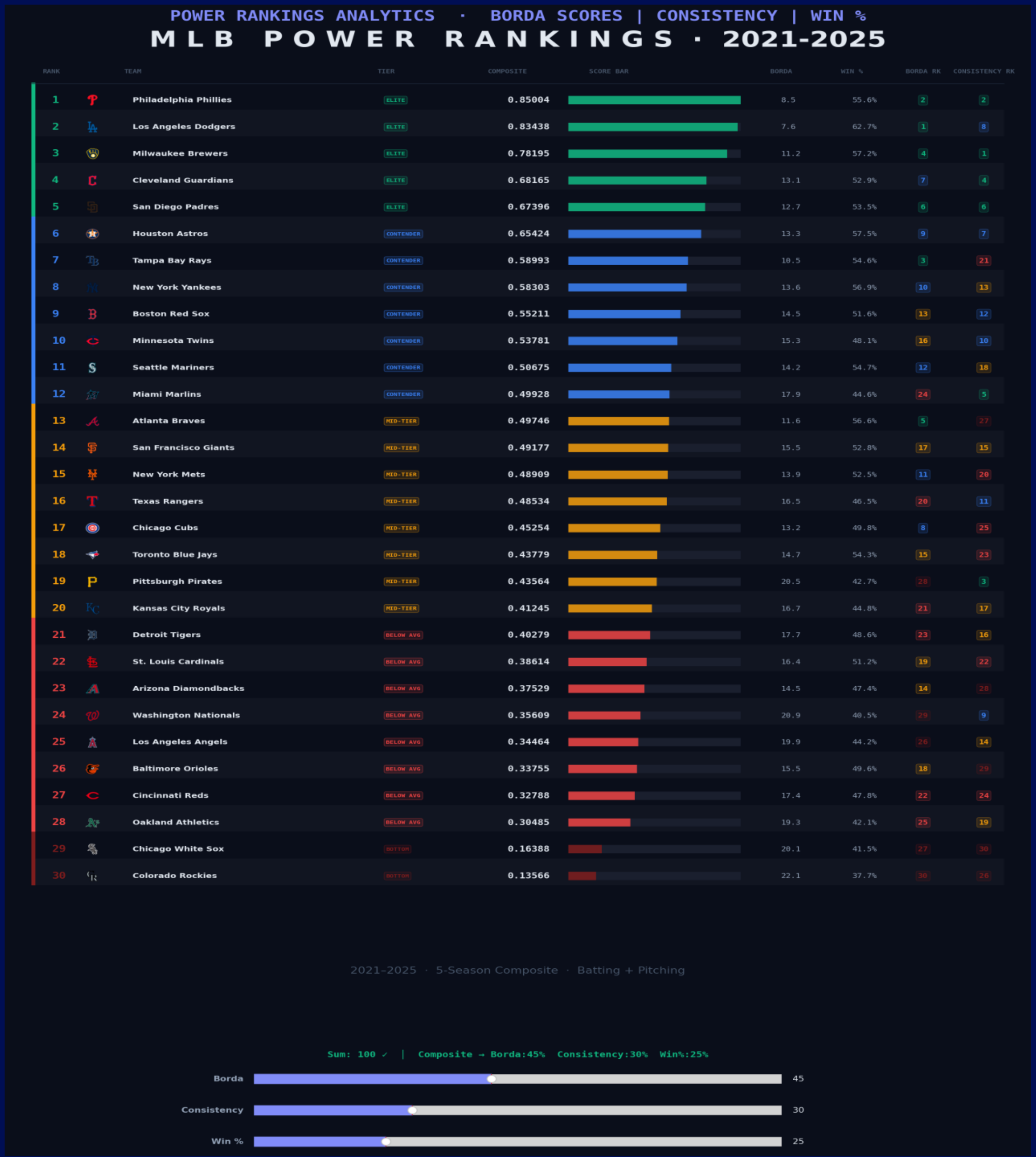
Top Ranked Teams:

Phillies (1), Dodgers (2), Brewers (3), Yankees (8)

The MLB Power Ranking derived from our borda scores consistently and accurately highlighted teams in both the top and bottom half of the MLB; lending validity to the results.

Bottom Ranked Teams:

Rockies (30), White Sox (29), Athletics (28), Angels (25)



POWER RANKINGS ANALYTICS · BORDA SCORES | CONSISTENCY | WIN %

MLB POWER RANKINGS · 2021-2025

RANK	TEAM	TIER	COMPOSITE	SCORE BAR	BORDA	WIN %	BORDA RK	CONSISTENCY RK
1	Philadelphia Phillies	ELITE	0.85004		8.5	55.6%	2	2
2	Los Angeles Dodgers	ELITE	0.83438		7.6	62.7%	1	8
3	Milwaukee Brewers	ELITE	0.78195		11.2	57.2%	4	1
4	Cleveland Guardians	ELITE	0.68165		13.1	52.9%	7	4
5	San Diego Padres	ELITE	0.67396		12.7	53.5%	6	6
6	Houston Astros	CONTENDER	0.65424		13.3	57.5%	9	7
7	Tampa Bay Rays	CONTENDER	0.58993		10.5	54.6%	3	21
8	New York Yankees	CONTENDER	0.58303		13.6	56.9%	10	13
9	Boston Red Sox	CONTENDER	0.55211		14.5	51.6%	13	12
10	Minnesota Twins	CONTENDER	0.53781		15.3	48.1%	16	10
11	Seattle Mariners	CONTENDER	0.50675		14.2	54.7%	12	18
12	Miami Marlins	CONTENDER	0.49928		17.9	44.6%	24	5
13	Atlanta Braves	MID-TIER	0.49746		11.6	56.6%	5	27
14	San Francisco Giants	MID-TIER	0.49177		15.5	52.8%	17	15
15	New York Mets	MID-TIER	0.48909		13.9	52.5%	11	20
16	Texas Rangers	MID-TIER	0.48534		16.5	46.5%	20	11
17	Chicago Cubs	MID-TIER	0.45254		13.2	49.8%	8	25
18	Toronto Blue Jays	MID-TIER	0.43779		14.7	54.3%	15	23
19	Pittsburgh Pirates	MID-TIER	0.43564		20.5	42.7%	28	3
20	Kansas City Royals	MID-TIER	0.41245		16.7	44.8%	21	17
21	Detroit Tigers	BELOW AVG	0.40279		17.7	48.6%	23	16
22	St. Louis Cardinals	BELOW AVG	0.38614		16.4	51.2%	19	22
23	Arizona Diamondbacks	BELOW AVG	0.37529		14.5	47.4%	14	28
24	Washington Nationals	BELOW AVG	0.35609		20.9	40.5%	29	9
25	Los Angeles Angels	BELOW AVG	0.34464		19.9	44.2%	26	14
26	Baltimore Orioles	BELOW AVG	0.33755		15.5	49.6%	18	29
27	Cincinnati Reds	BELOW AVG	0.32788		17.4	47.8%	22	24
28	Oakland Athletics	BELOW AVG	0.30485		19.3	42.1%	25	19
29	Chicago White Sox	BOTTOM	0.16388		20.1	41.5%	27	30
30	Colorado Rockies	BOTTOM	0.13566		22.1	37.7%	30	26

2021–2025 · 5-Season Composite · Batting + Pitching

2025 Season - Best MLB Teams

2025 Philadelphia Phillies Statistics

« 2024 Season   2026 Season »

via Sports Logos.net
About logos

Record: 96-66, 1st place in NL East (Schedule and Results)
Postseason: Lost NL Division Series (3-1) to Los Angeles Dodgers
Manager: Rob Thomson (96-66)
President: David Dombrowski (President, Baseball Operations)
General Manager: Preston Mattingly (VP & GM)
Farm Director: Luke Murton
Scouting Director: Brian Barber

2025 Milwaukee Brewers Statistics

« 2024 Season   2026 Season »

via Sports Logos.net
About logos

Record: 97-65, 1st place in NL Central (Schedule and Results)
Postseason:
Lost NL Championship Series (4-0) to Los Angeles Dodgers
Won NL Division Series (3-2) over Chicago Cubs

2025 Los Angeles Dodgers Statistics

« 2024 Season   2026 Season »

via Sports Logos.net
About logos

Record: 93-69, 1st place in NL West (Schedule and Results)
Postseason:
Won World Series (4-3) over Toronto Blue Jays
Won NL Championship Series (4-0) over Milwaukee Brewers
Won NL Division Series (3-1) over Philadelphia Phillies
Won NL Wild Card Series (2-0) over Cincinnati Reds

Hitting Pitching

2025 | Regular Season | MLB | Select a Split | Standard | Expan

	TEAM	LEAGUE	G	AB	R	H	2B	3B	HR	RBI	BB	SO	SB	CS	AVG	OBP
1	Toronto Blue Jays	AL	162	5507	798	1461	294	13	191	771	520	1099	77	25	.265	.333
2	Milwaukee Brewers	NL	162	5510	806	1423	265	18	166	750	564	1266	164	53	.258	.332
3	New York Yankees	AL	162	5471	849	1371	255	20	274	820	639	1463	134	38	.251	.332
4	Philadelphia Phillies	NL	162	5517	778	1426	268	24	212	753	528	1337	124	30	.258	.328
5	Los Angeles Dodgers	NL	162	5481	825	1384	257	21	244	791	580	1353	88	25	.253	.327

Hitting Pitching

2025 | Regular Season | MLB | Select a Split | Standard | Expande

	TEAM	LEAGUE	W	L	ERA	G	GS	CG	SHO	SV	SVO	IP	H	R	ER	HR	HB	BB	SO
1	Los Angeles Dodgers	NL	93	69	3.95	162	162	0	10	46	73	1441.0	1250	683	633	175	65	563	1505
2	Houston Astros	AL	87	75	3.86	162	162	3	12	45	62	1442.0	1257	665	619	199	58	508	1504
3	Philadelphia Phillies	NL	96	66	3.79	162	162	2	14	47	74	1440.1	1342	648	607	177	52	435	1471
4	New York Yankees	AL	94	68	3.91	162	162	0	11	43	64	1439.2	1239	685	625	175	57	557	1440
5	Milwaukee Brewers	NL	97	65	3.58	162	162	0	12	45	66	1442.0	1237	634	574	168	66	534	1432
6	Toronto Blue Jays	AL	94	68	4.19	162	162	2	8	42	65	1438.0	1313	721	669	209	72	517	1430

Hitting Pitching

2025 | Regular Season | MLB | Select a Split

	TEAM	LEAGUE	G	AB	R
1	New York Yankees	AL	162	5471	849
2	Los Angeles Dodgers	NL	162	5481	825
3	Milwaukee Brewers	NL	162	5510	806
4	Toronto Blue Jays	AL	162	5507	798
5	Chicago Cubs	NL	162	5495	793
6	Arizona Diamondbacks	NL	162	5480	791
7	Boston Red Sox	AL	162	5562	786
8	Philadelphia Phillies	NL	162	5517	778

2025 Season - Worst MLB Teams

via Sports Logos.net
About logos

2025 Los Angeles Angels Statistics

« 2024 Season | 2026 Season »

Record: 72-90, 5th place in AL_West (Schedule and Results)
Managers: Ron Washington (36-38) and Ray Montgomery (36-52)
General Manager: Perry Minasian
Farm Director: Joey Prebynski

via Sports Logos.net
About logos

2025 Colorado Rockies Statistics

« 2024 Season | 2026 Season »

Record: 43-119, 5th place in NL_West (Schedule and Results)
Managers: Bud Black (7-33) and Warren Schaeffer (36-86)
General Manager: Bill Schmidt (Senior VP & GM)
Farm Director: Chris Forbes

via Sports Logos.net
About logos

2025 Chicago White Sox Statistics

« 2024 Season | 2026 Season »

Record: 60-102, 5th place in AL_Central (Schedule and Results)
Manager: Will Venable (60-102)
General Manager: Chris Getz (Senior VP & GM)
Farm Director: Paul Janish

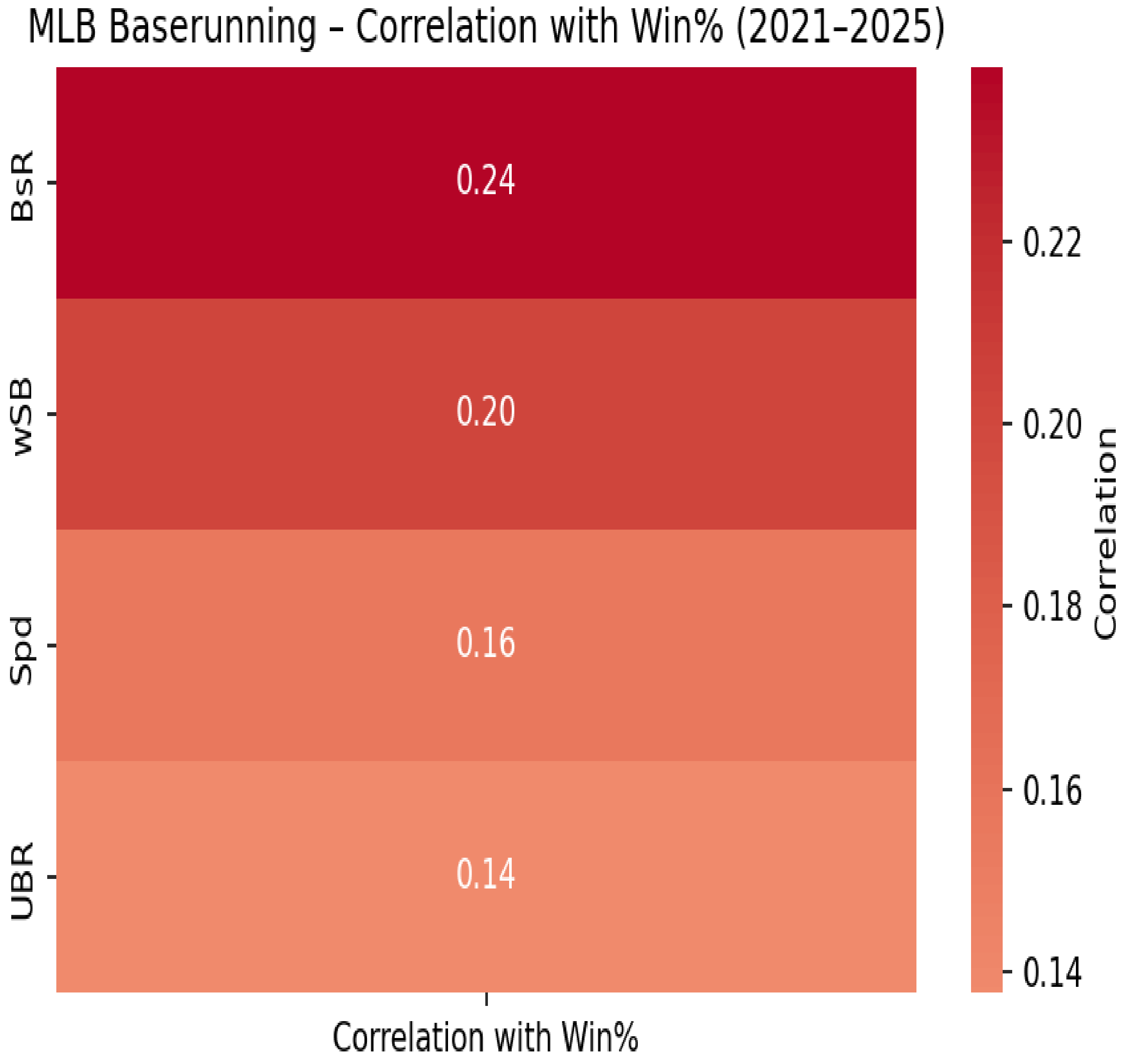
	TEAM	LEAGUE	G	AB	R	H	2B	3B	HR	RBI	BB	SO	SB	CS	AVG	OBP
1	Colorado Rockies	NL	162	5408	597	1281	253	37	160	581	395	1531	87	39	.237	.293
2	Cleveland Guardians	AL	162	5310	643	1199	243	18	168	621	494	1344	129	34	.226	.296
3	Los Angeles Angels	AL	162	5374	673	1209	212	17	226	649	484	1627	88	27	.225	.298
4	Chicago White Sox	AL	162	5377	647	1250	243	10	165	626	498	1364	85	21	.232	.302

2025 | Regular Season | MLB | Select a Split

	TEAM	LEAGUE	G	AB	R	H	2B	3B	HR	RBI	BB	SO
1	Los Angeles Angels	AL	162	5374	673	1209	212	17	226	649	484	1627
2	Colorado Rockies	NL	162	5408	597	1281	253	37	160	581	395	1531
3	New York Yankees	AL	162	5471	849	1371	255	20	274	820	639	1463
4	Baltimore Orioles	AL	162	5416	677	1273	251	19	191	643	484	1457
5	Detroit Tigers	AL	162	5456	758	1346	244	36	198	724	511	1454
6	Seattle Mariners	AL	162	5502	766	1345	234	9	238	734	544	1446
7	Pittsburgh Pirates	NL	162	5375	583	1244	245	21	117	561	530	1422
8	Boston Red Sox	AL	162	5562	786	1414	324	24	186	748	518	1419
9	Cincinnati Reds	NL	162	5448	716	1333	250	23	167	677	527	1415
10	Athletics	AL	162	5547	733	1403	296	16	219	709	502	1406

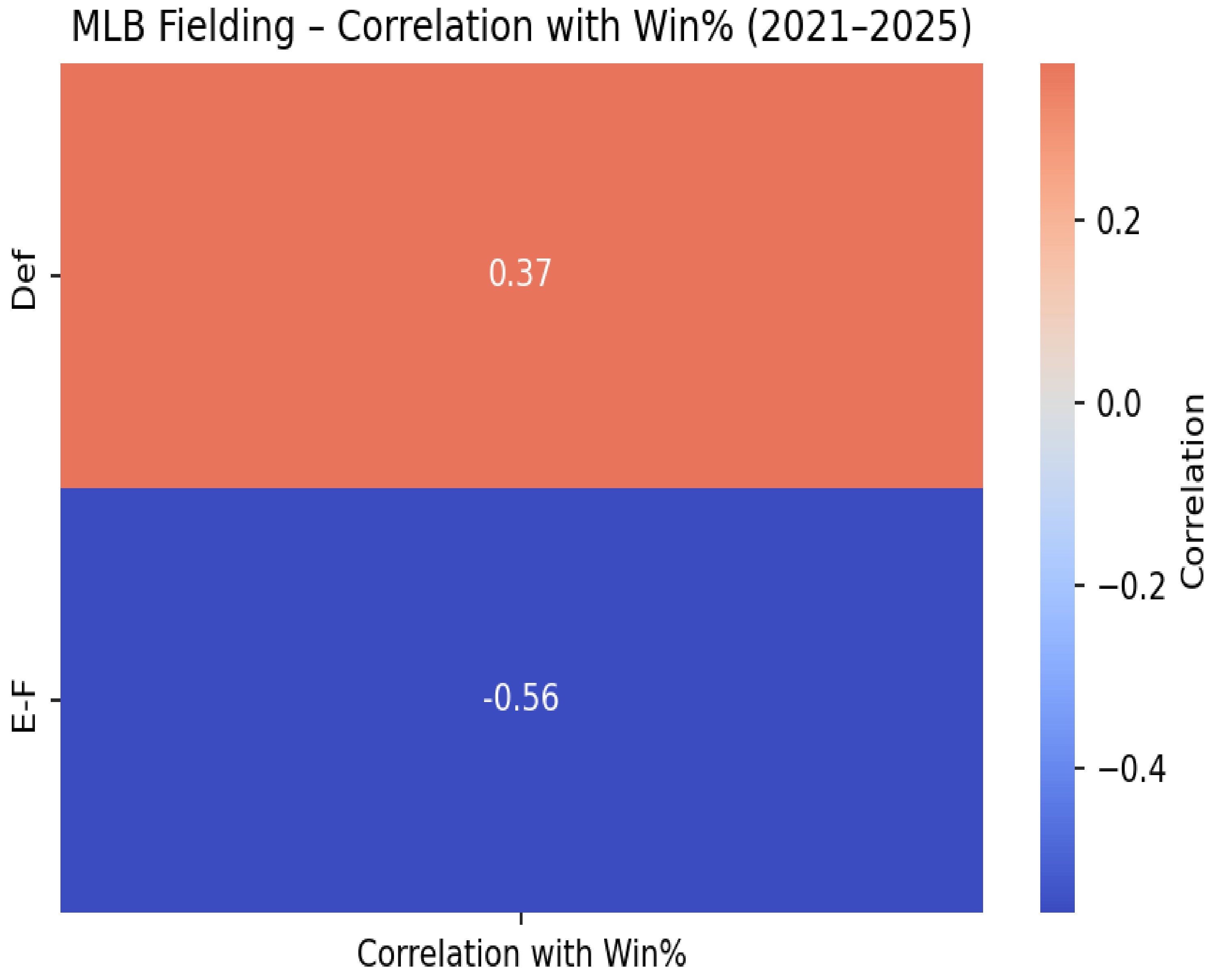
2025 | Regular Season | MLB | Select a Split

	TEAM	LEAGUE	W	L	ERA
1	Colorado Rockies	NL	43	119	5.97
2	Washington Nationals	NL	66	96	5.35
3	Los Angeles Angels	AL	72	90	4.89
4	Athletics	AL	76	86	4.70
5	Baltimore Orioles	AL	75	87	4.60



Given that I used team hitting and pitching graphs, the baserunning and fielding observations were always going to be ancillary.

That being said, BsR yields a decent correlation to winning (24%). Obviously, it is always a positive when good baserunning results in extra run scoring opportunities.



I think the most significant observation that can be taken from the analysis of fielding is that poor fielding seems to contribute more to a team losing than good fielding does to help a team win.

Similar to pitching, fielding acquisitions may be informed by avoidance of poor fielders as opposed to seeking out good ones.

MLB Conclusion

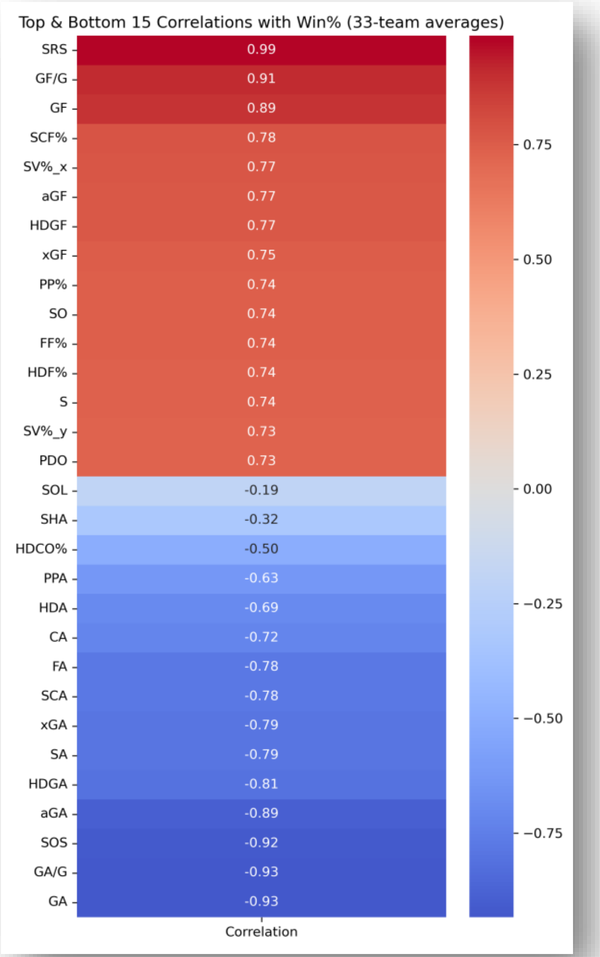
- Creating additional scoring opportunities (wRC+, OBP, and BsR). reliably in offensive scenarios (hitting and baserunning) increased odds of winning.
- Defensive Metrics (Pitching and Fielding) yielded stronger results on the negatively correlated side, highlighting potential performance results to avoid (ERA, WHIP, and FIP).

Overall, as is the case in most sports; scoring more points results in more wins while allowing points ends in losing. As evidenced by the gathered data, the consequences of playing negative players is more significant than acquiring positive players.

NHL OVERVIEW

All Data Gathered from Hockey-Reference.com

NHL Positive / Negative Correlations: Heatmap



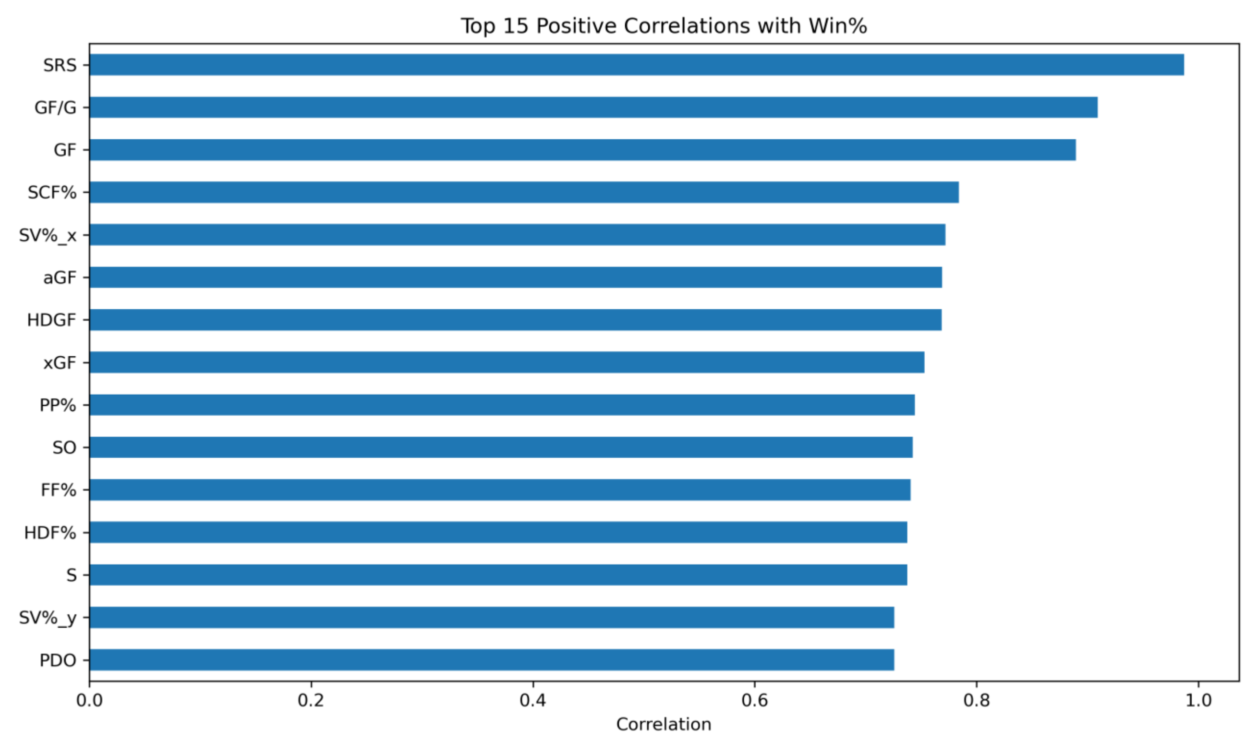
Seasons Reviewed: 2021—2025

Overview:

• Displays the correlation between every team-level statistic and win percentage, ranked from strongest positive to strongest negative
• Deep red at the top and deep blue at the bottom show the relationships are not subtle
• Stats like SRS, GF/G, and SCF% sit firmly in positive territory, while GA, SOS, and aGA sit firmly in the negative
• Gives a quick, color-coded snapshot of which metrics matter most when predicting long-term NHL success across a 5-year window

Based on 2025 data shown in picture, other seasons may have some variance

NHL Positive Correlations: Overview



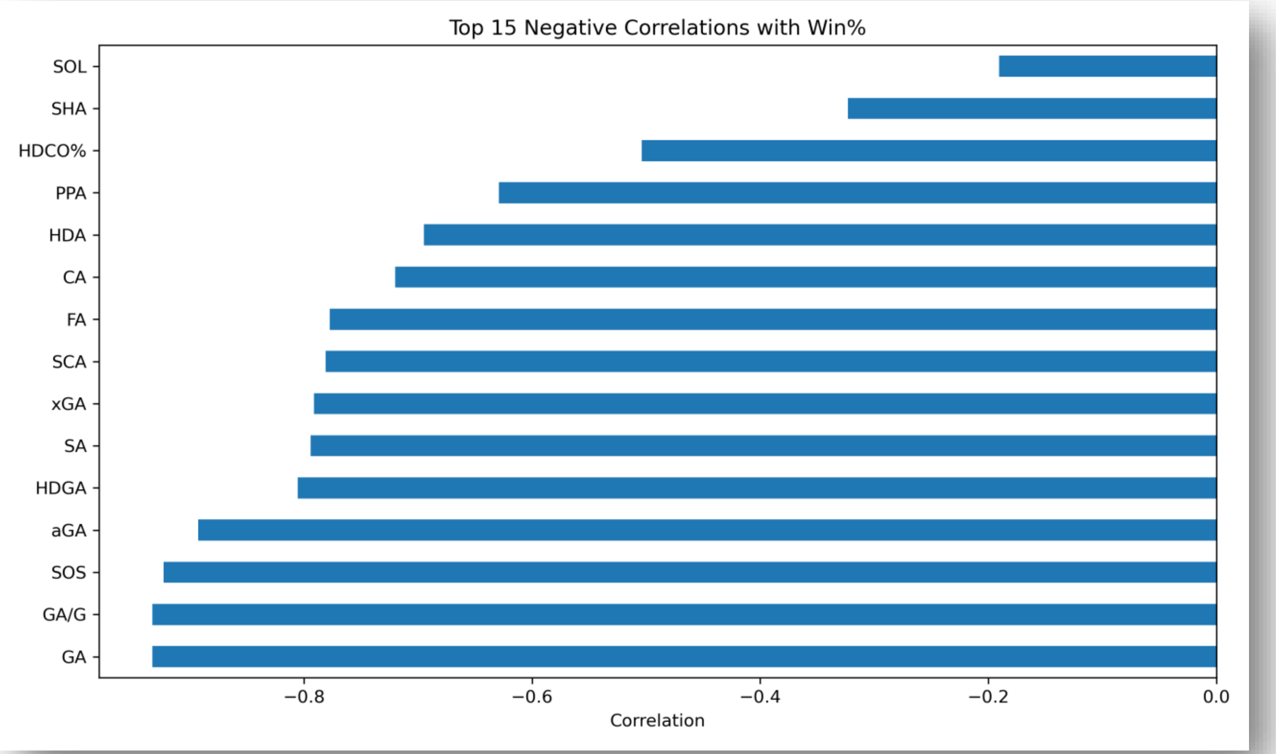
Seasons Reviewed: 2020—2024

Overview:

• Strongest positive correlates with win percentage were SRS at 0.99, GF/G at 0.91, and GF at 0.89, confirming that scoring volume and scoring margin are the clearest separators between franchises over time
• SCF% at 0.78 stood out as the most analytically significant finding
• Scoring Chance For Percentage is a process metric that measures territorial control in dangerous areas, independent of whether shots actually go in
• Its strong correlation suggests that teams who consistently dominate play in high-danger zones win over a 5-year span even after individual game luck evens out, validating the direction modern NHL analytics has taken

Based on 2025 data shown in picture, other seasons may have some variance

NHL Negative Correlations: Overview



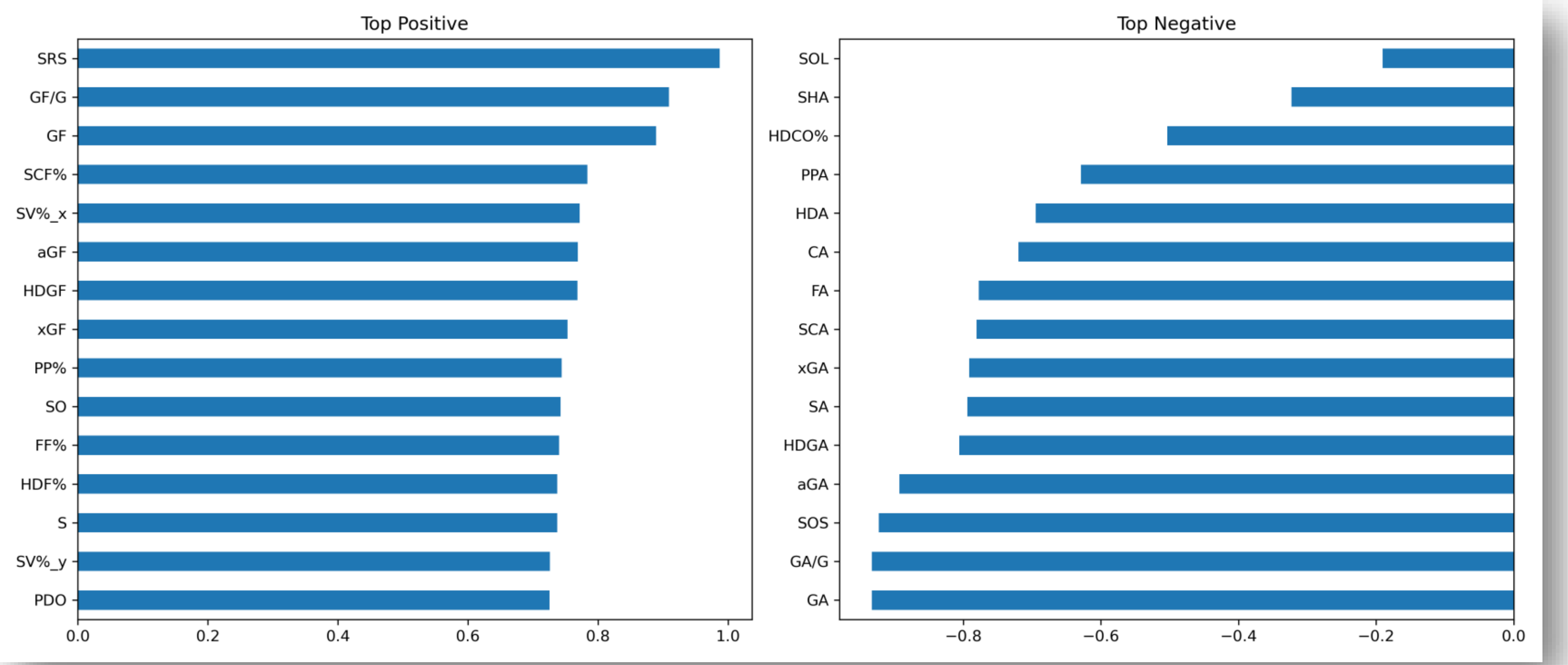
Seasons Reviews: 2020—2024

Findings:

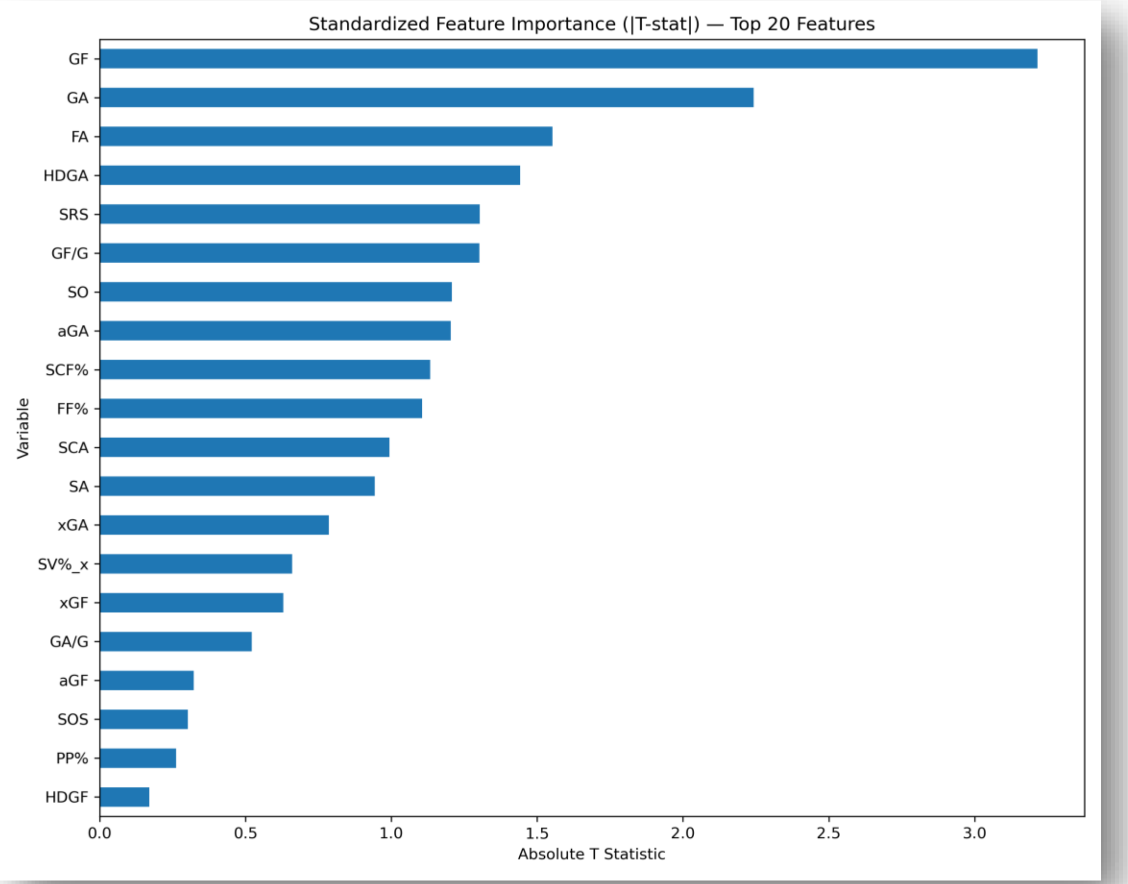
- Goals Against and Goals Against per Game both correlated at -0.93 with win percentage, nearly matching the strength of the positive scoring metrics on the other side
- Confirms that defense and offense contribute almost equally to long-term success in the NHL, and that no team can simply outscore its problems over five seasons
- Strength of Schedule came in at -0.92, which at first seems counterintuitive but actually reflects structural imbalances in the NHL schedule
- Teams in weaker divisions naturally accumulate more wins over a full era regardless of true quality

Based on 2025 data shown in picture, other seasons may have some variance

NHL Positive and Negative Correlations: Comparison



NHL Standarized Feature Importance



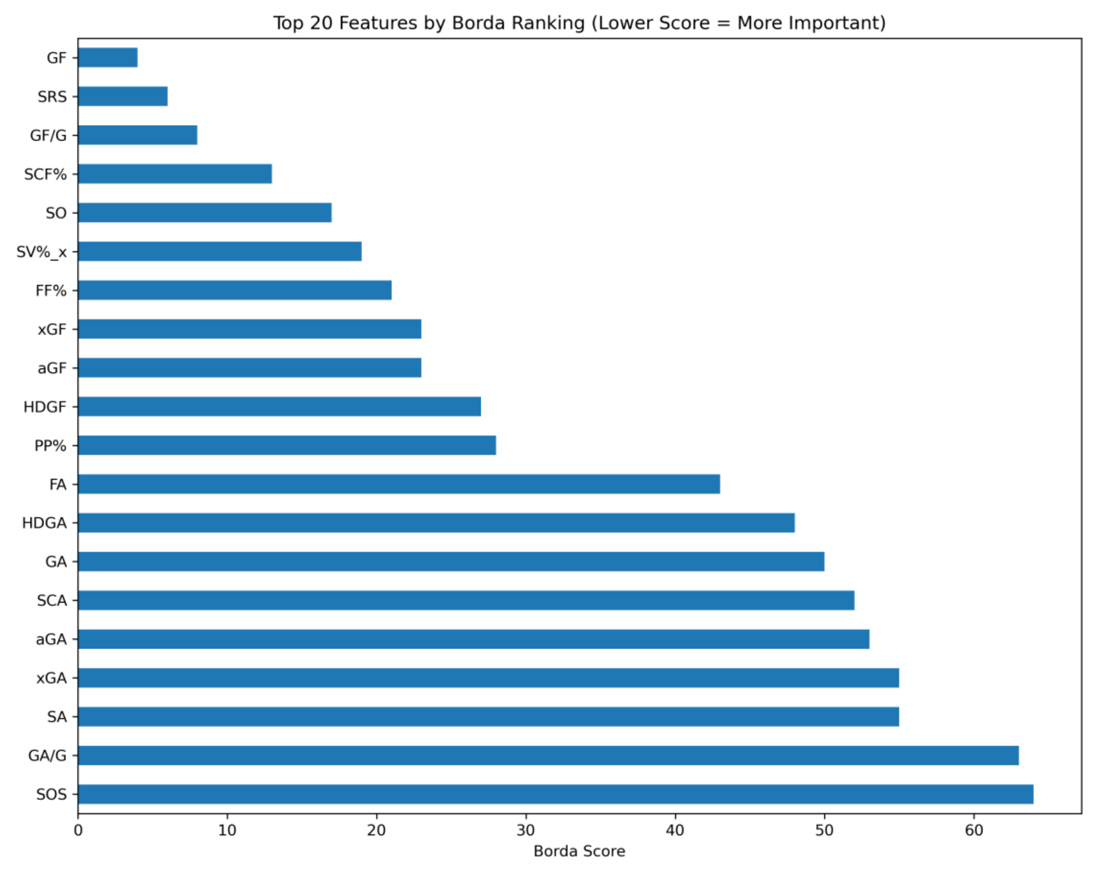
Seasons Reviewed: 2020—2024

Overview:

- Runs OLS regression on the top 20 features after scaling all variables to the same unit, making bar lengths directly comparable across metrics of different scales
- Gives a cleaner picture of which features have the strongest independent effect on win percentage after controlling for everything else
- Unlike raw correlation, which only measures the one-to-one relationship between a stat and winning, this approach shows which metrics carry unique explanatory power once the others are accounted for
- AvAge and PDO rise to the top here, a more nuanced finding that raw correlation alone would never have surfaced

Based on 2025 data shown in picture, other seasons may have some variance

NHL Top Features by Borda Ranking



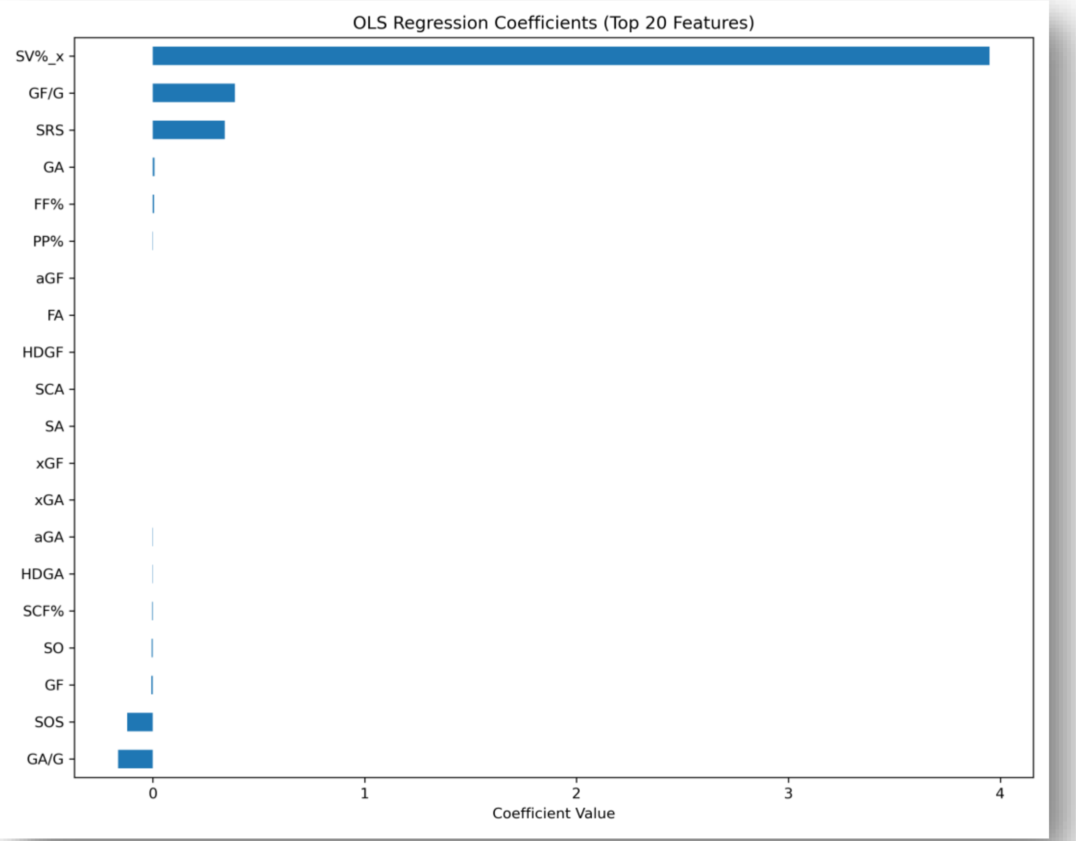
Seasons Reviewed: 2020—2024

Findings:

• Combined two independent methods, correlation rank and regression T-stat rank, into a single importance score where lower means more consistently important across both approaches
• GF came out on top with a score of 4.0, followed by SRS at 6.0 and GF/G at 8.0, with SCF% and Shutouts rounding out the top five
• Most notable result is that GF ranked above SRS despite SRS having the higher raw correlation
• Once other variables were controlled for in the regression, raw goals scored had a stronger independent signal, identifying scoring volume as the single most separating factor between franchises over a 5-year window

Based on 2024 data shown in picture, other seasons may have some variance

NHL OLS Regression



Seasons Reviewed: 2020—2024

Overview:

- Run on the top 20 features selected by absolute correlation with win percentage, a necessary step given that 33 observations and 48 available features would otherwise produce perfect multicollinearity and invalid results
- Produced an R-squared of 0.992, meaning the selected metrics collectively explain 99.2% of the variation in win percentage across the 33 franchises
- F-statistic p-value of 1.10e-09 confirms the model is statistically significant as a whole
- Individual coefficients show some multicollinearity given the small sample size, which is why the Borda and standardized importance methods are used alongside OLS rather than relying on any single approach alone

Based on 2025 data shown in picture, other seasons may have some variance

NHL Borda: Power Rankings

Seasons Reviewed: 2020—2024

Talking Points:

- **Florida Panthers:** The Panthers have won the previous two Stanley Cup Finals and have some hockey commentators and media personalities asking if they have become a "dynasty". Yet, their Borda score and consistency ranking do not even place them into the elite category. This furthered our group's discussion of the inherent difference between the regular season and the playoffs.
- **A between-league comparison:** as discussed previously, would be the Buffalo Bills, who have been dominant in the regular season but have never reached a Super Bowl. The similarities between leagues are perfectly highlighted by comparing the Buffalo Bills to the Carolina Panthers, who ranked #1 in Borda and #3 in consistency, but like the Bills, have been unable to reach a Stanley Cup final over the past five seasons.
- **Boston Bruins:** The traditionally strong Bruins have shown they can compete with the best in the NHL over the last 5 seasons; however, they have lacked consistency in their statistical output to truly challenge the top of the league. This is best highlighted by the Bruins having 1st- or 2nd-round exits in the four seasons prior to missing the playoffs last season.
- **Edmonton Oilers:** A team that has reached the previous two Stanley Cup Finals but ranked #11 in Borda across the previous 5 seasons. This may represent a team on the rise. One area that may warrant further investigation is the seasonal decay factor, which could change or lessen the impact of older seasons.

Conclusions

- Teams' statistical performances show many similarities and differences across leagues when comparing stats to win percentage. We found that, although strong offensive stats are important, defense also matters! There are likely confounding factors influencing statistical performance; as discussed in the presentation, financial constraints or the lack thereof vary from league to league. While the MLB and EPL do not have salary caps, the NBA has a soft cap, which is inherently different from the NFL’s hard cap.

- We found that using Borda Scores was an efficient and effective way to compare teams, both within and across leagues, to assess how they performed relative to their statistical outputs.

- Finally, consistency across seasons varies by league, with different levels of influence in the leagues we evaluated. Again, confounding factors are likely to have an unmeasured influence and would require further research in future projects. Roster sizes, cap structures, and league financial structures affect how a team can build a roster. It is clear from our research that certain statistical profiles matter more than others, but outside factors must be addressed before drawing any definitive conclusions.